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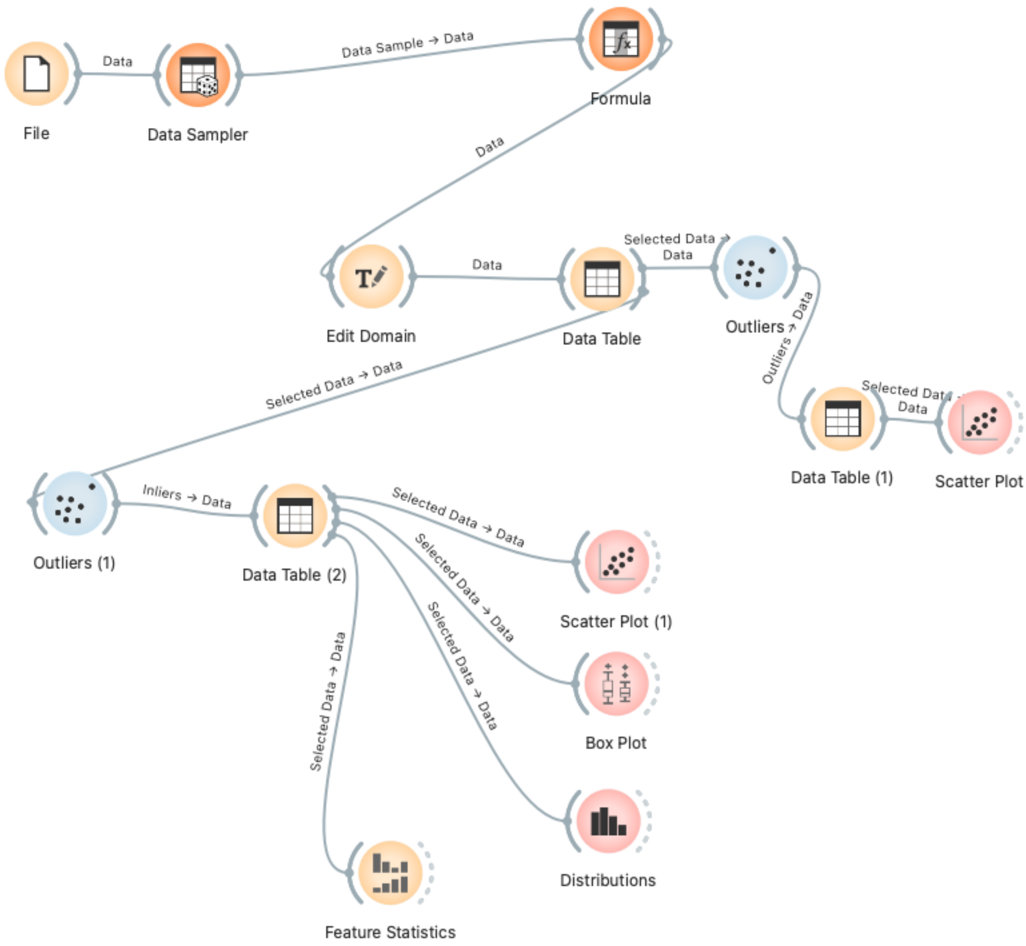
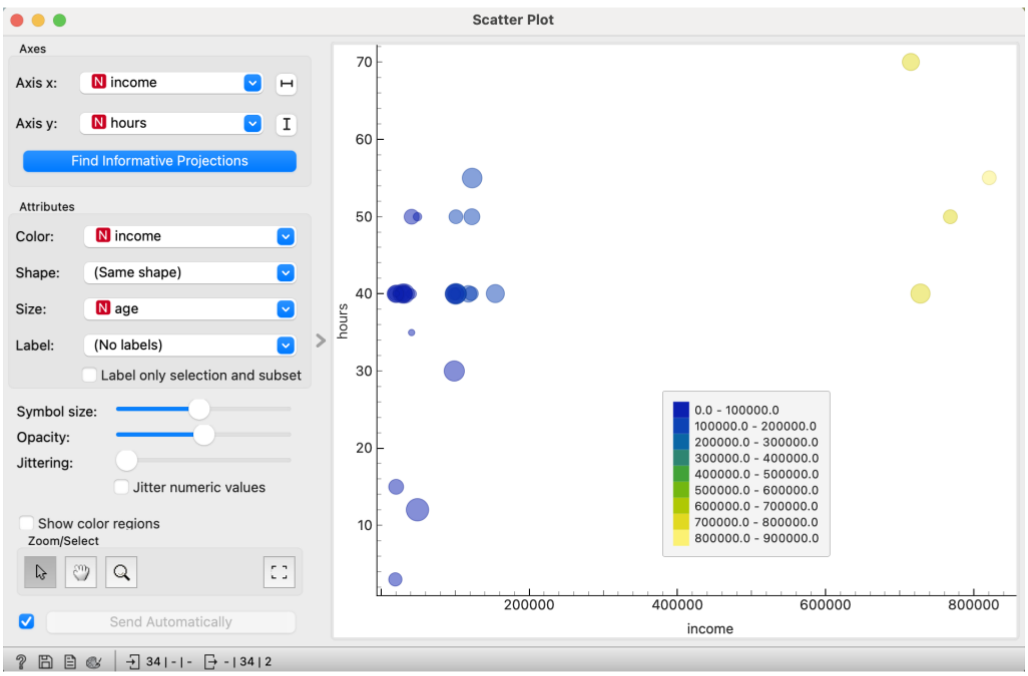
Executive Summary

This data mining report provides a comprehensive analysis of income disparities across various demographics, utilising a range of statistical methods and visualisations. The report begins with a detailed preprocessing phase, where a random sampler was used to select 5,000 data points to manage computational load while ensuring a representative sample. Visualisations such as scatter plots and box plots were employed to identify and manage outliers, which are critical for accurate analysis. The analysis reveals significant income disparities across sex, place of birth, and race, with males, US-born, and "White" individuals generally earning more. The report also explores the relationship between income and other variables such as age, hours worked, and education, finding weak positive correlations that are strengthened by log transformations. Classification models were employed to predict income categories, with logistic regression and Naive Bayes identified as the most effective models. Feature ranking identified income, sex, and race as significant predictors. Further on, the report will analyse voting data, examining different factors influencing political affiliations, using maps and other graphs. This analysis aims to provide insights into how demographic factors influence voting behaviour, with a focus on the 2020 U.S. presidential election.

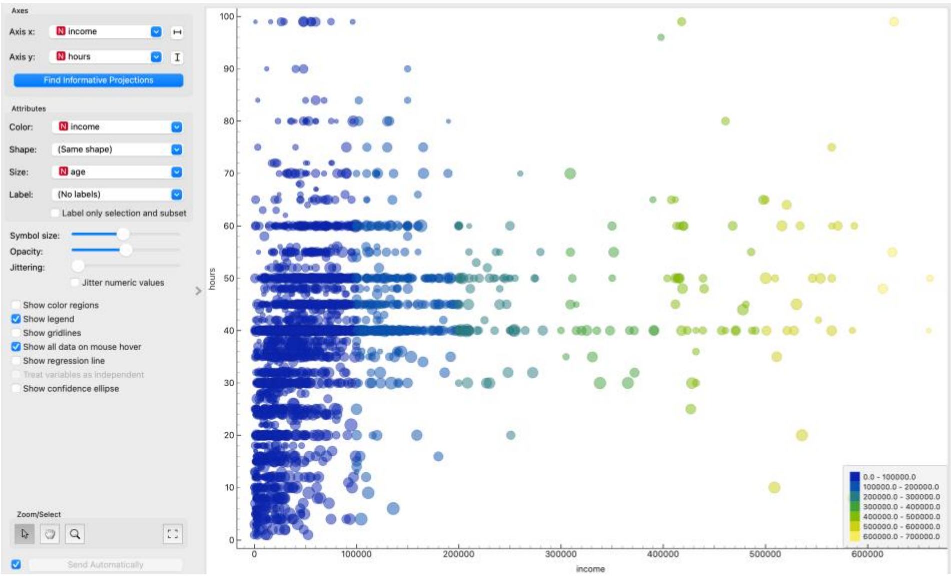
Part 1- Preprocessing



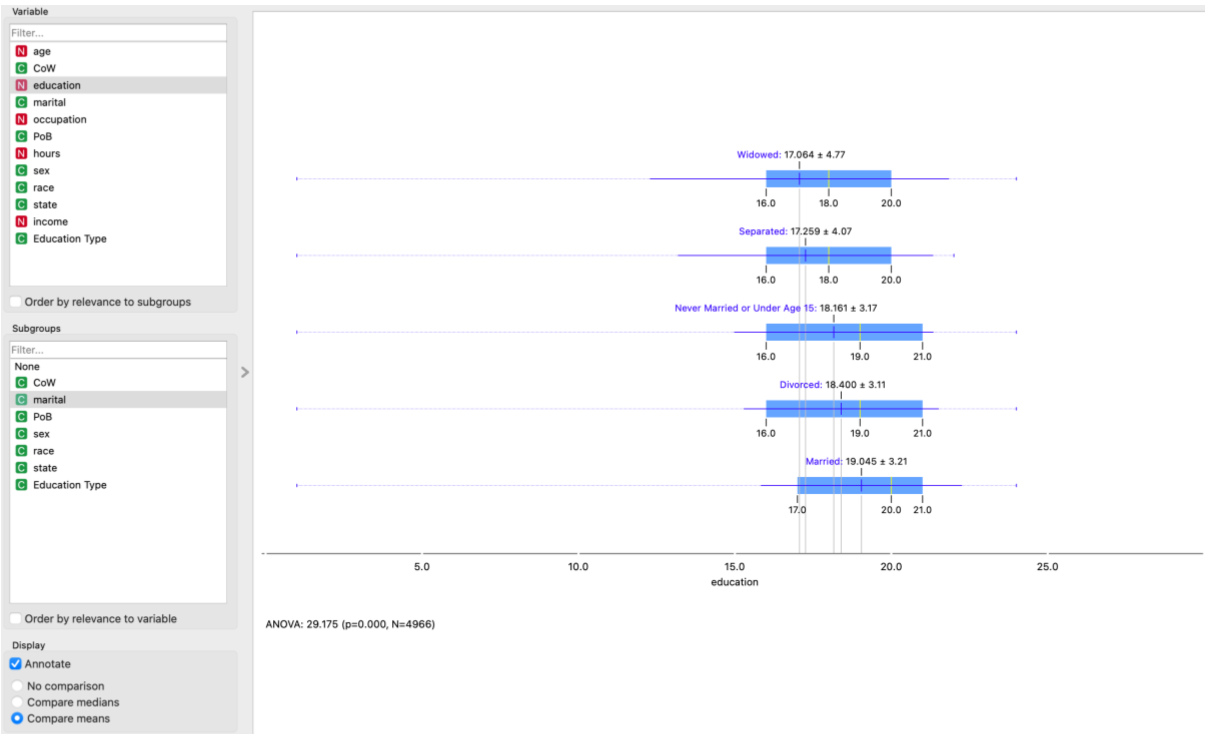
Outliers Scatterplot



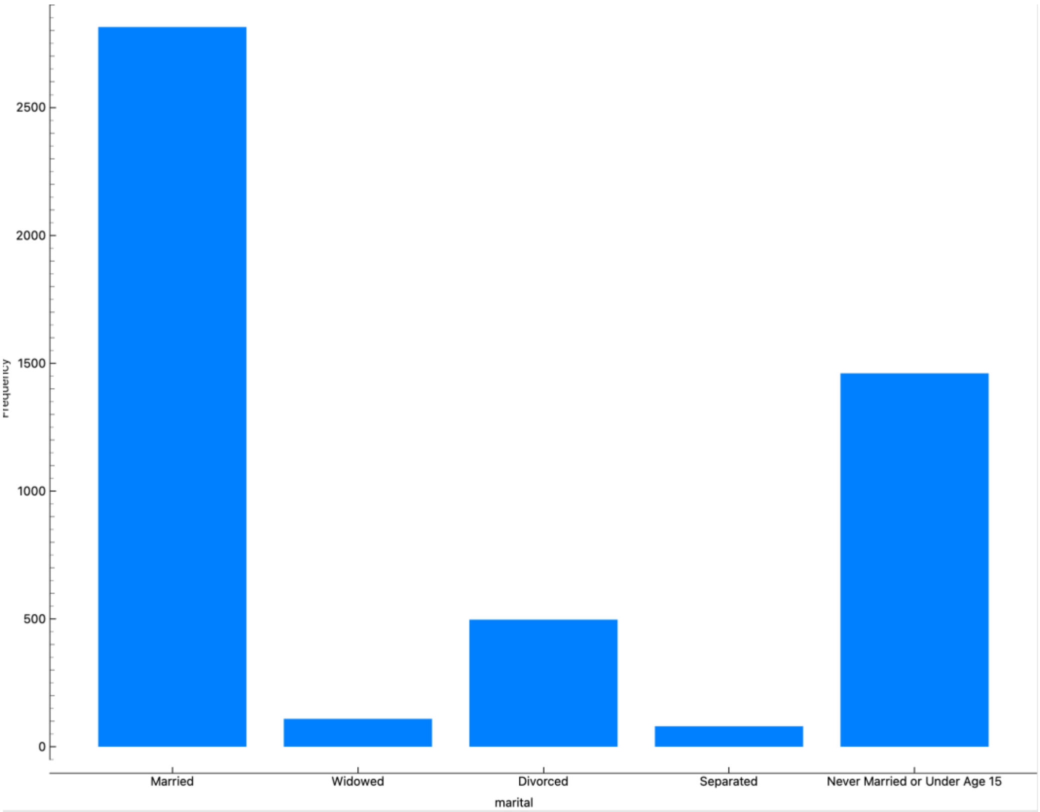
Inliers Scatterplot



Box Plots



Distributions



Feature Statistics

	Name	Distribution	Mean	Mode	Median	Dispersion	Min.	Max.	Missing
N	age		43.676	57.0	44.0	0.348	17.0	95.0	0 (0 %)
N	education		18.629	21.0	19.0	0.174	1.0	24.0	0 (0 %)
N	occupation		4182.967	2310.0	4220.0	0.638	10.0	9830.0	0 (0 %)
N	PoB		66.077	6.0	36.0	1.403	1.0	501.0	0 (0 %)
N	hours		38.456	40.0	40.0	0.344	1.0	99.0	0 (0 %)
N	income		56828.873	50000.0	40000.0	1.217	120.0	690000.0	0 (0 %)
C	CoW			Employee of a private for-profit company or business, or ...		1.27			0 (0 %)
C	marital			Married		1.06			0 (0 %)
C	sex			Male		0.692			0 (0 %)
C	race			White		0.519			0 (0 %)

Analysis

To ensure the accuracy and reliability of the results in this coursework analysis, several preprocessing steps were essential. Given the extensive size of the dataset, I employed a random sampler to select 5,000 data points for analysis. This was crucial to manage computational load while ensuring that the sample was representative of the entire dataset, thereby enhancing the generalisability of the findings.

I used the edit domain widget to restructure the data, standardising the format and ensuring consistency across the dataset, which is critical for accurate analysis. To correctly name the various attributes, I referred to the attribute_values Excel document. This step was vital for correctly labelling the attributes, ensuring accurate interpretation of results, and maintaining data integrity.

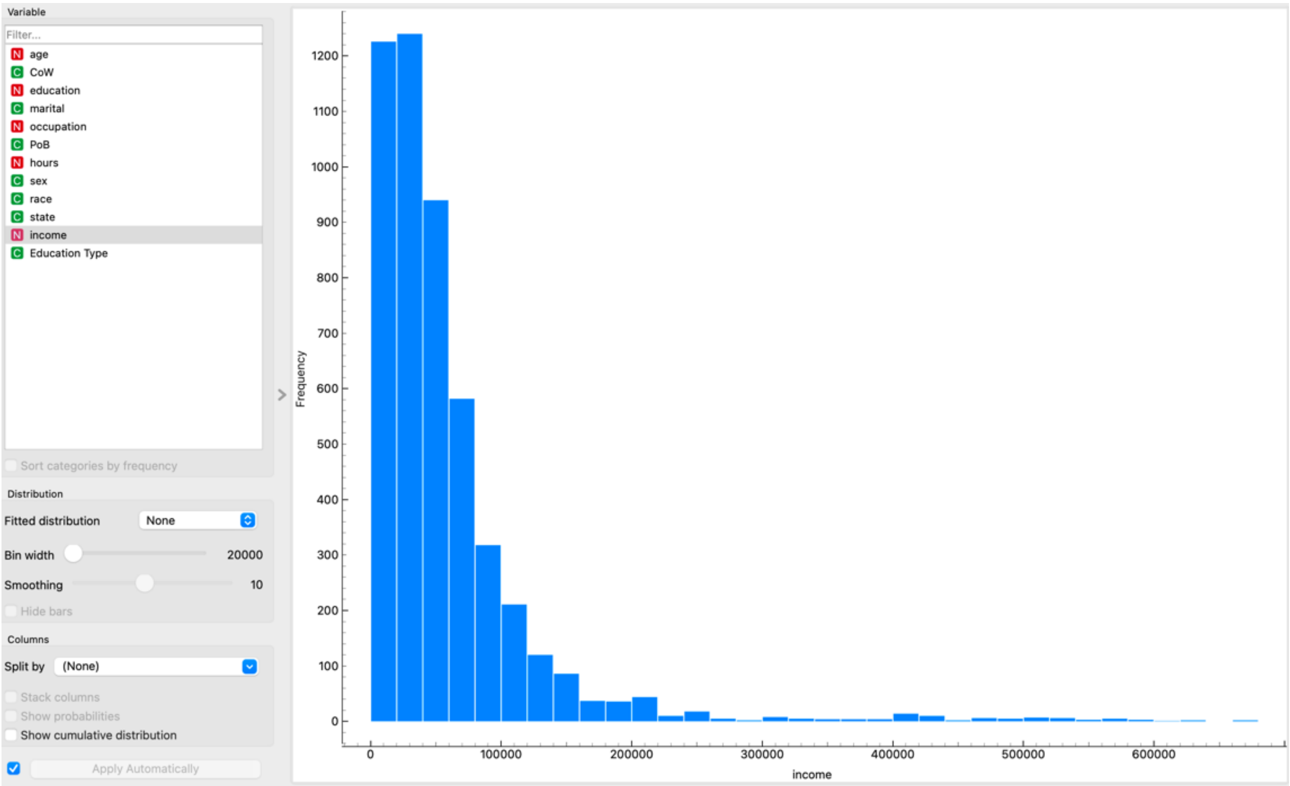
For the occupation, place of birth, and education data, I used the formula widget before the edit domain to duplicate these attributes, maintaining one version as numeric and the other as text for editing purposes. This dual representation provided flexibility in analysis, accommodating different data types required by various analytical methods. I then connected the data to a data table to verify the correct format before proceeding. This verification step was crucial to prevent errors in subsequent analyses.

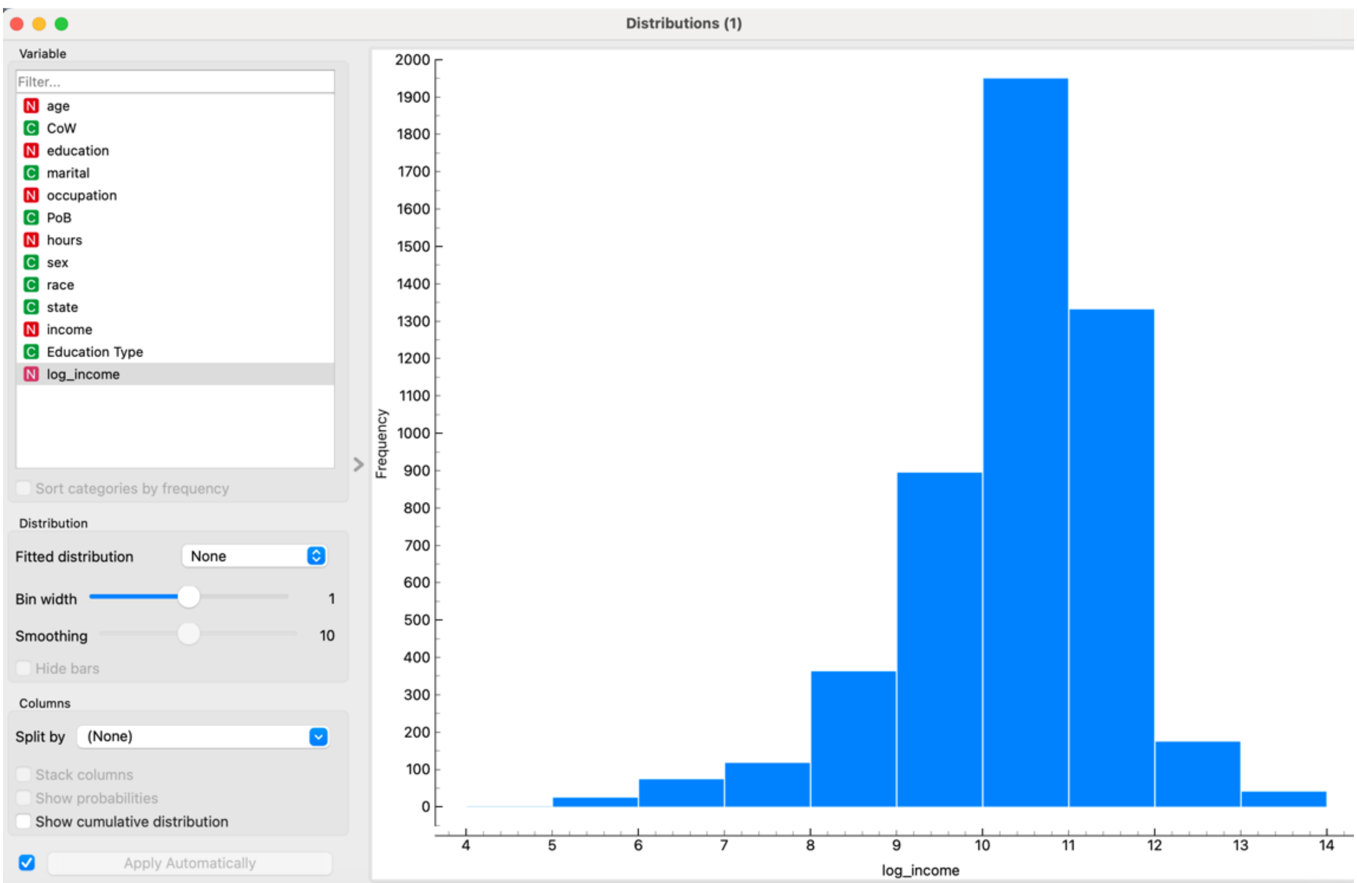
To address outliers, I incorporated two outlier widgets—one for outliers and one for inliers—from the data table. These were visualised using a scatter plot. Identifying and managing outliers is critical as they can significantly skew results and lead to misleading conclusions. By focusing on inliers, I ensured that the analysis was based on the most representative data points, thereby enhancing the validity of the findings. The outlier scatterplot revealed a few particularly high values in the right-hand corner; removing these outliers will lead to more accurate analysis. For the rest of my analysis, I will be using the inliers.

To gain a comprehensive understanding of the dataset's structure and key characteristics, I examined key statistics from the feature statistics, scatterplots, distributions, and box plots. I adjusted the labels within the graphs to facilitate clearer understanding of significant visualisations and statistics. This step was important to ensure correct data interpretation and accurate identification of patterns or trends. This also allowed me to understand any potential bias in the data, such as with the majority of this data set being from married people, as seen by the distribution graph. By thoroughly understanding the dataset, I was able to make informed decisions throughout the analysis process, which is essential for producing reliable and meaningful results.

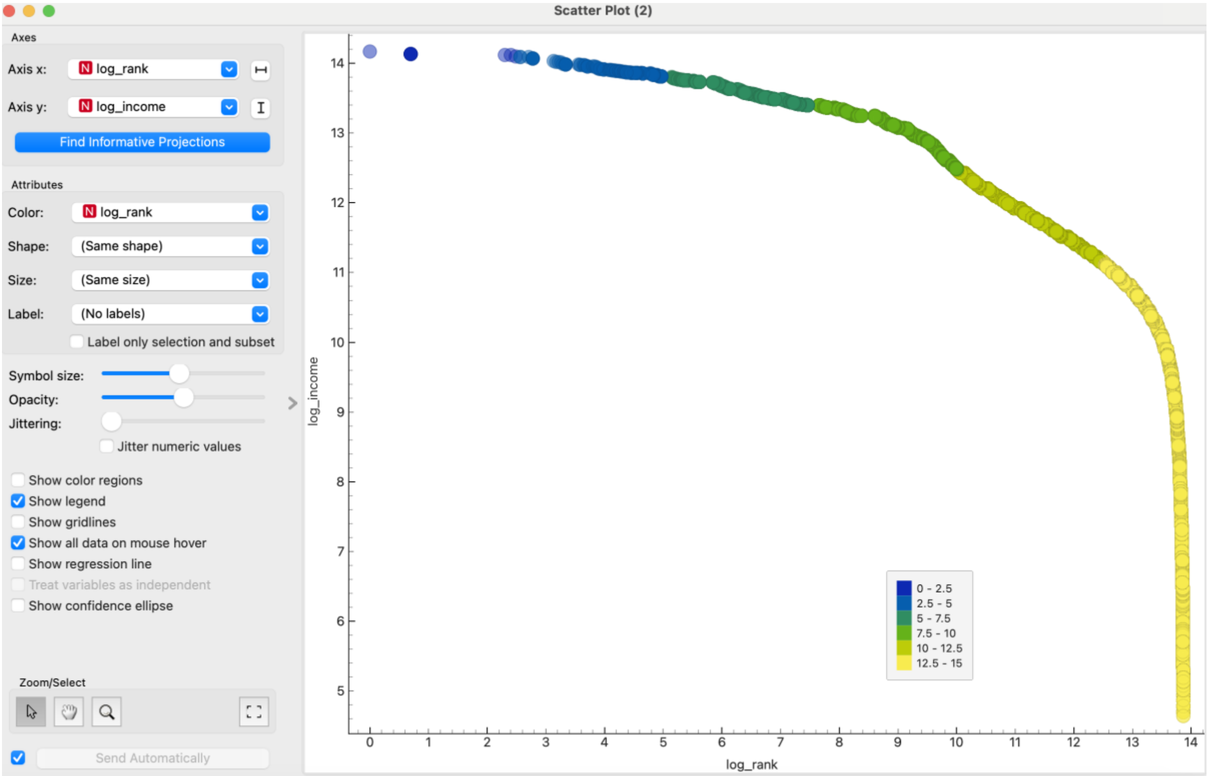
Part 2- Fairness in Income Distribution

Histogram of income / log income





Zipt plot for income

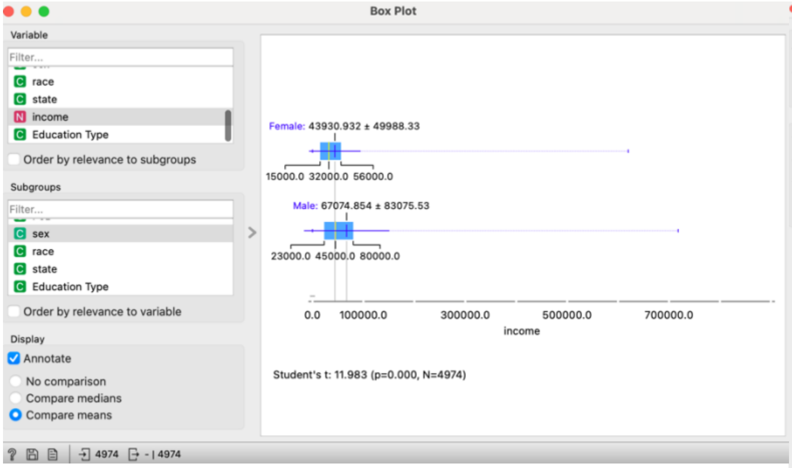


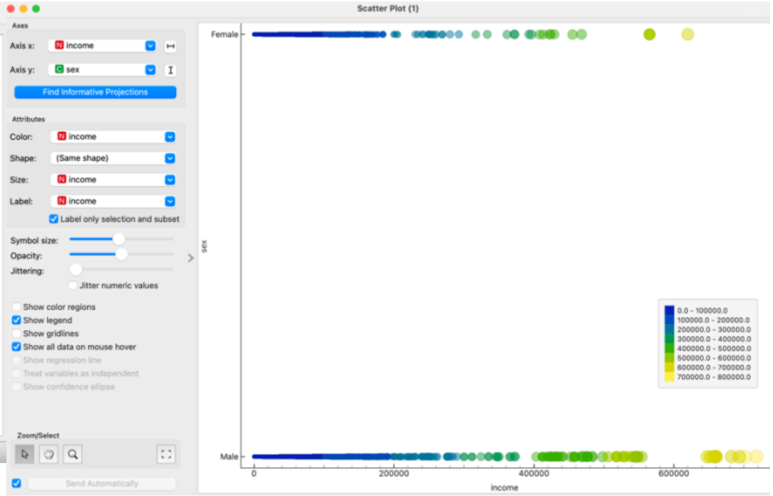
The income histogram shows a skewed distribution, with most people earning low incomes and fewer people earning high incomes, suggesting significant income inequality. This pattern might result from varying education levels, occupational differences, work experience disparities, and systemic factors favouring high earners. The log income graph, a histogram of income logarithms, resembles a slightly right-skewed normal distribution, cantered around a log income of 10. This indicates most individuals have log incomes near this value. The log transformation compresses high-income data, reducing outlier impact and revealing a more normal distribution.

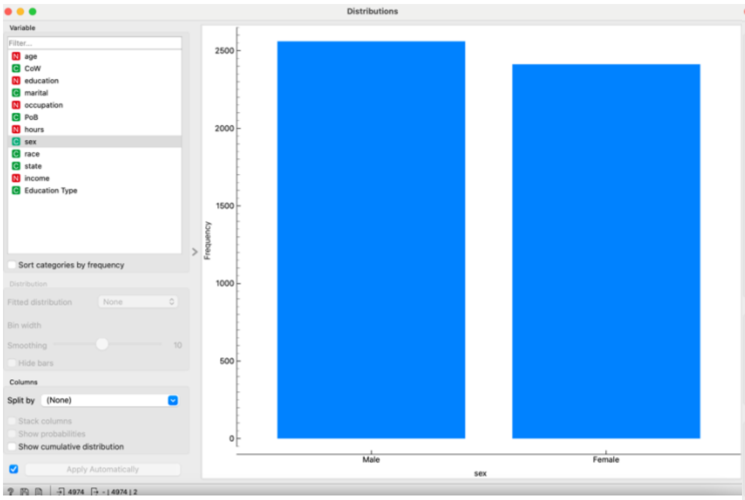
The Zipf plot shows an inverse relationship between log rank and log income, consistent with Zipf's law, highlighting that many individuals fall into the lower income brackets, whereas only a few occupy the higher income ranges. This graph also provides a more normalised distribution.

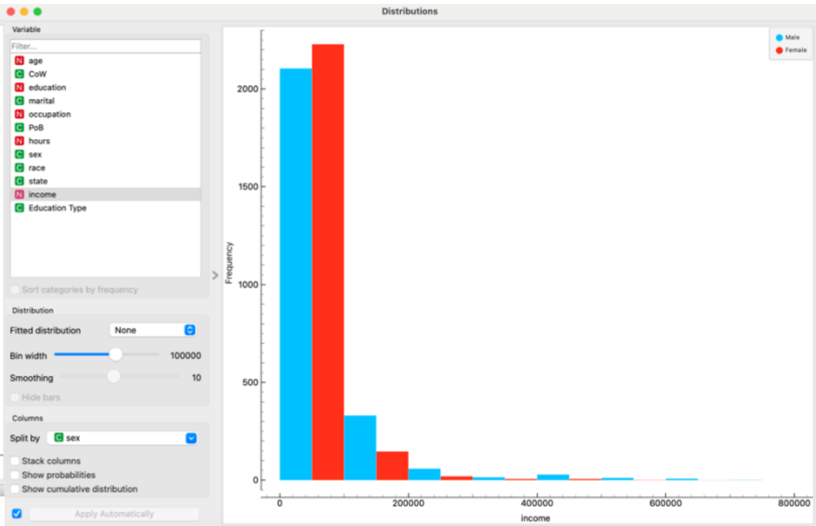
Regarding demographics, a box plot shows males have a higher median income and more high-income outliers than females. A Student's t-test ($t = 11.93$, $p < 0.001$, $N = 4974$) confirms higher mean income for males. For place of birth, US-born individuals have slightly higher median incomes, but a t-test ($t = 0.840$, $p = 0.401$, $N = 4974$) shows no significant mean income difference. Race analysis reveals "White" individuals have higher median incomes and more high-income outliers, supported by a t-test ($t = 5.004$, $p < 0.001$, $N = 4974$). However, potential biases and dataset imbalances must be considered, and further analysis is needed for robust conclusions.

Sex against income graphs

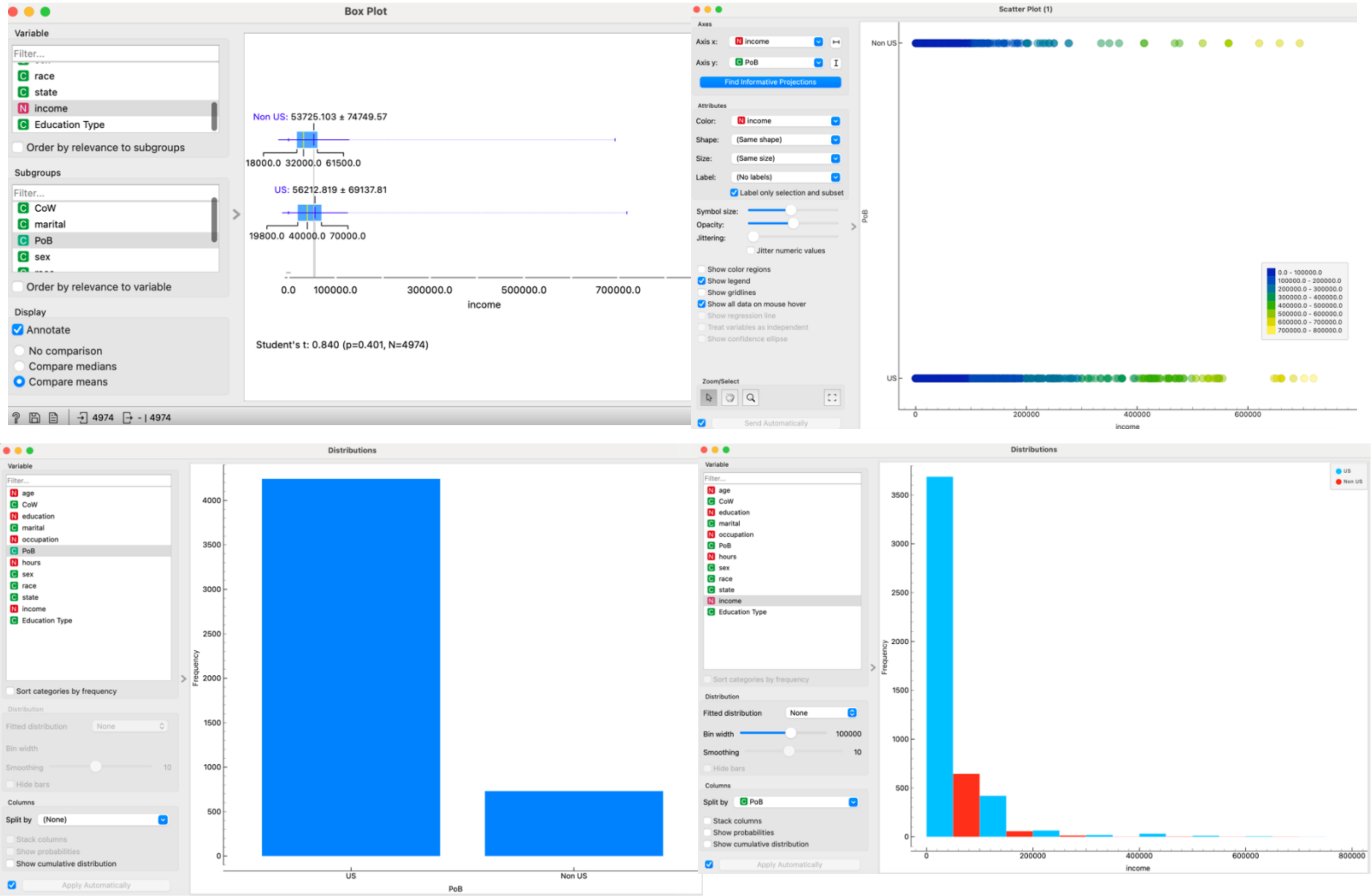




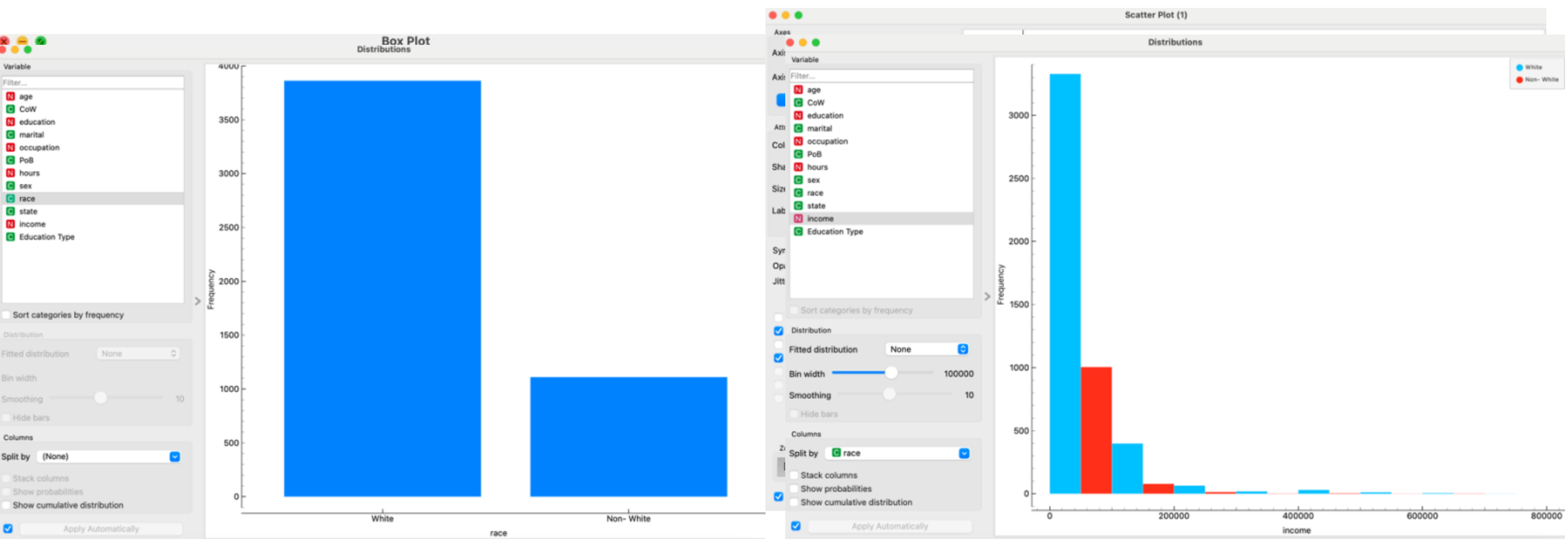




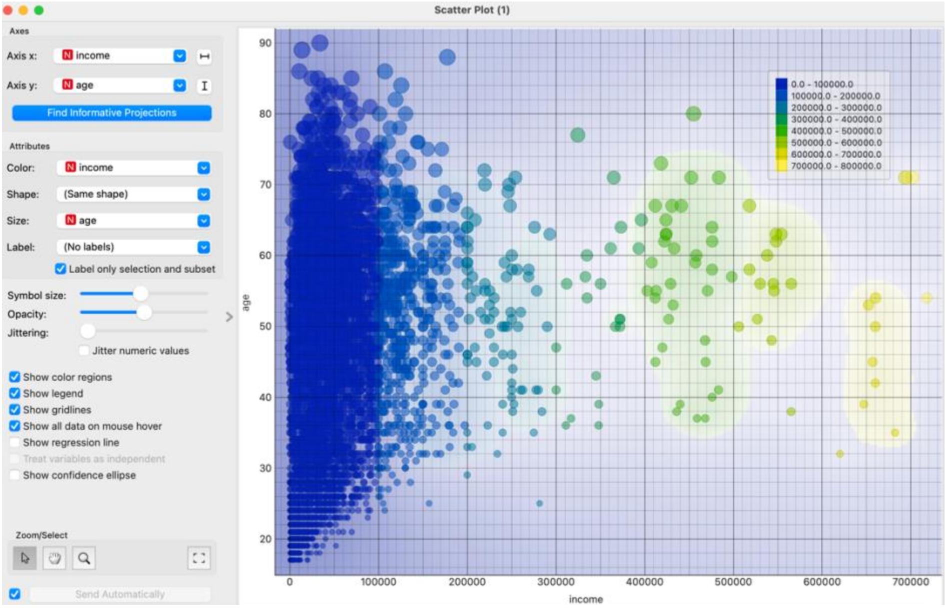
Place of birth against income graphs

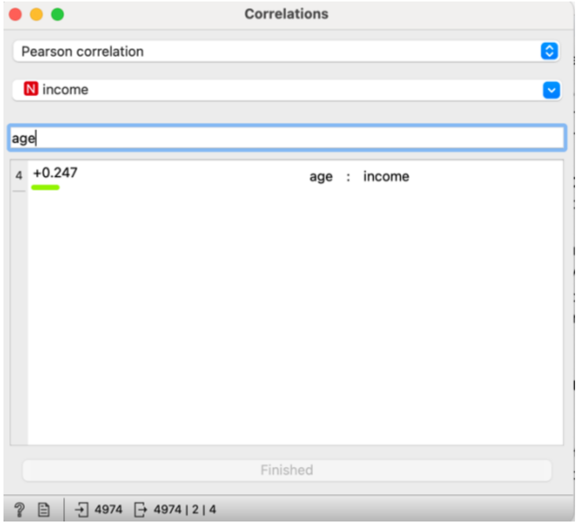


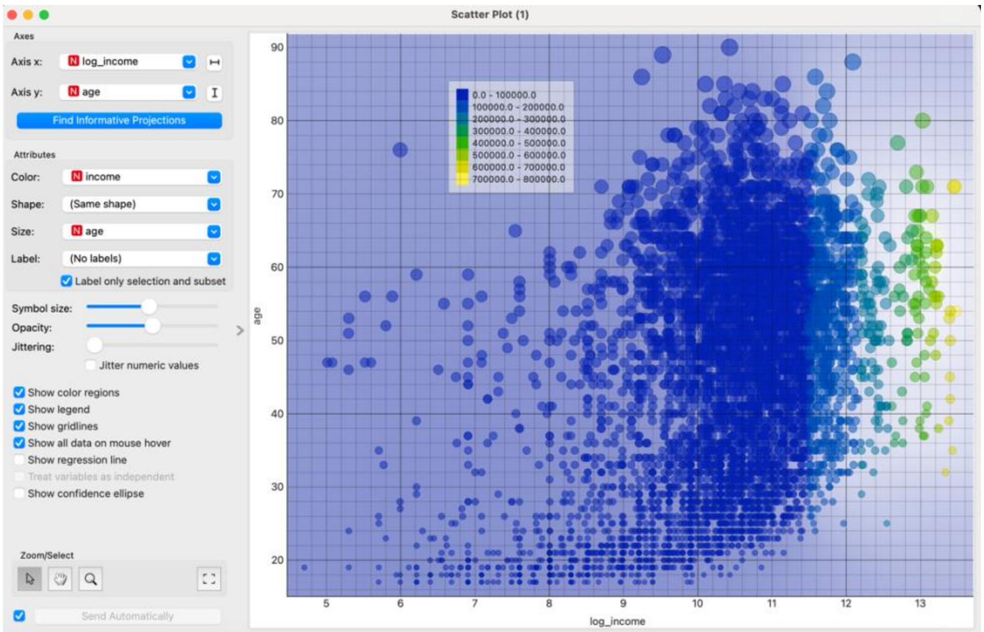
Race against income graphs

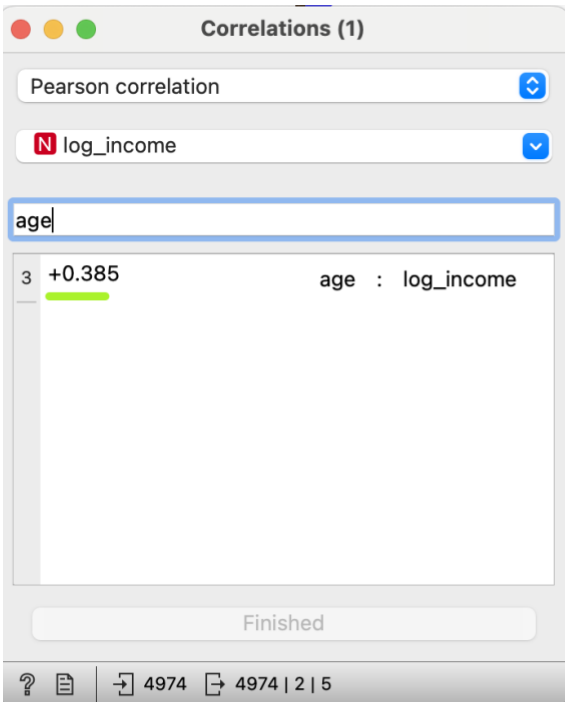


Income and age graphs

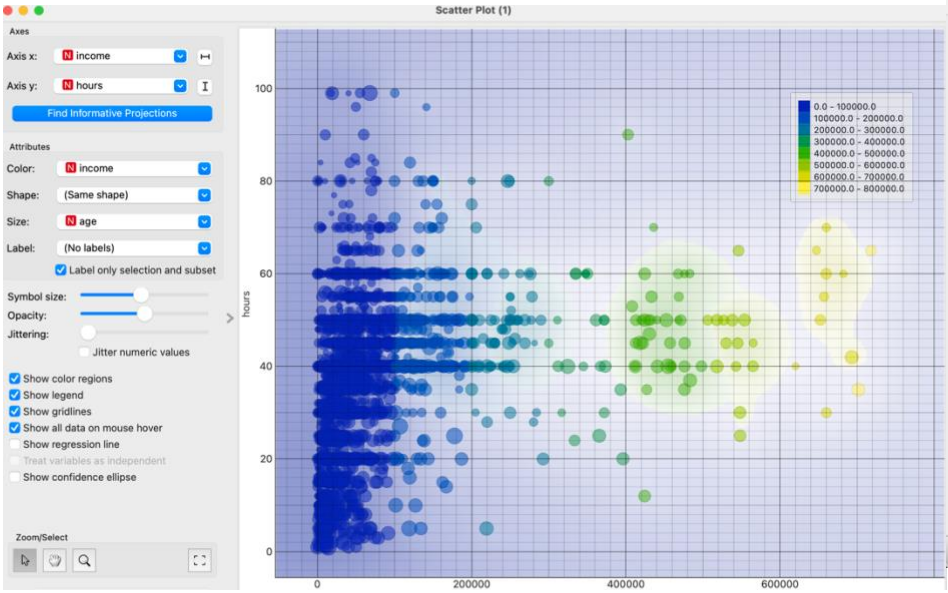


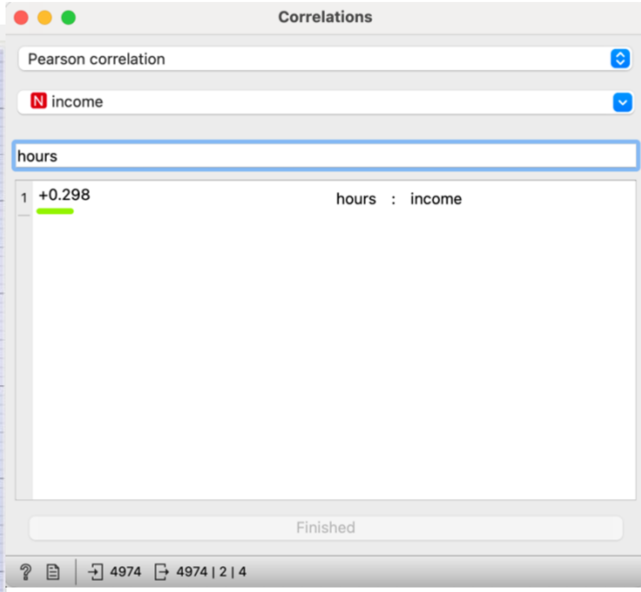


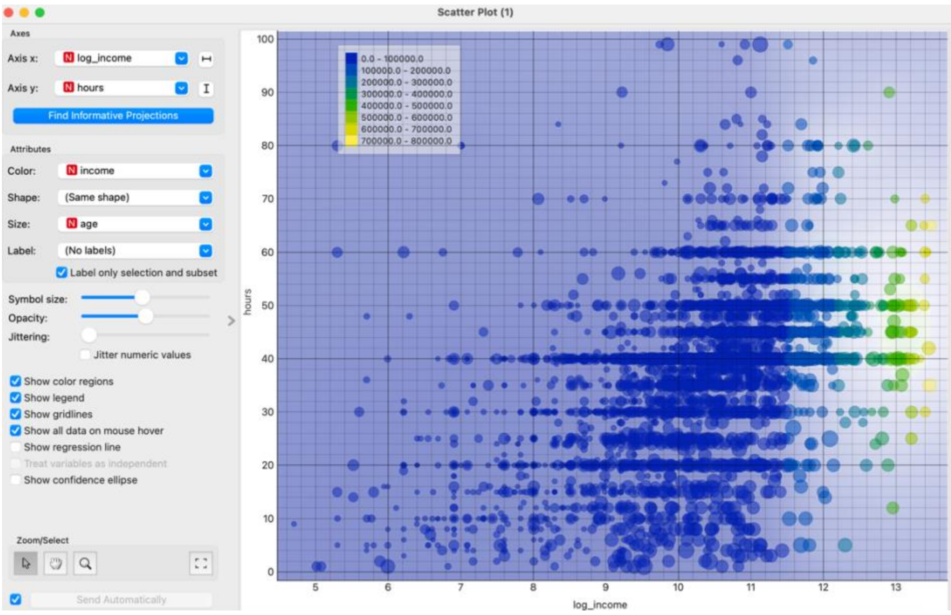


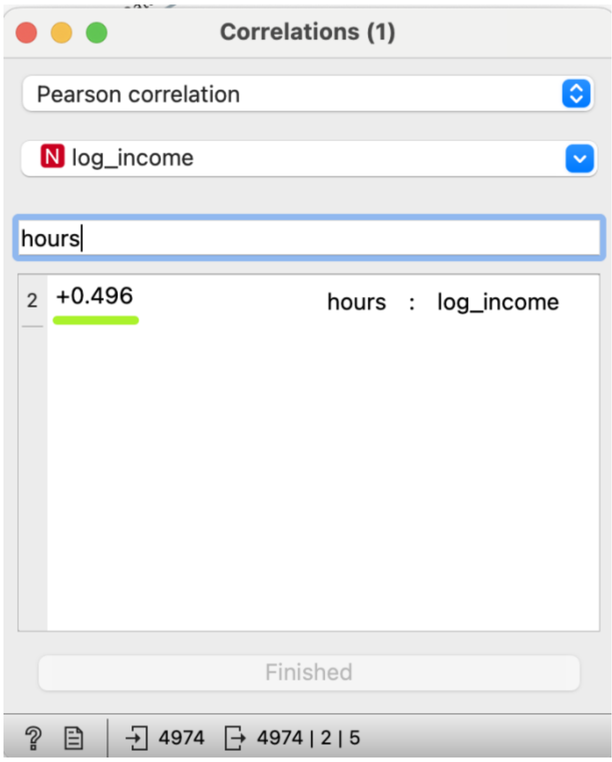


Income and hours worked per week graphs

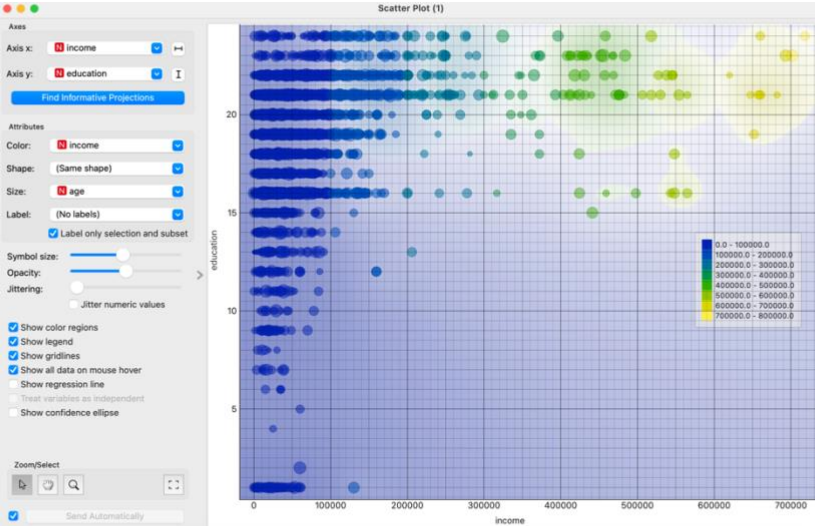


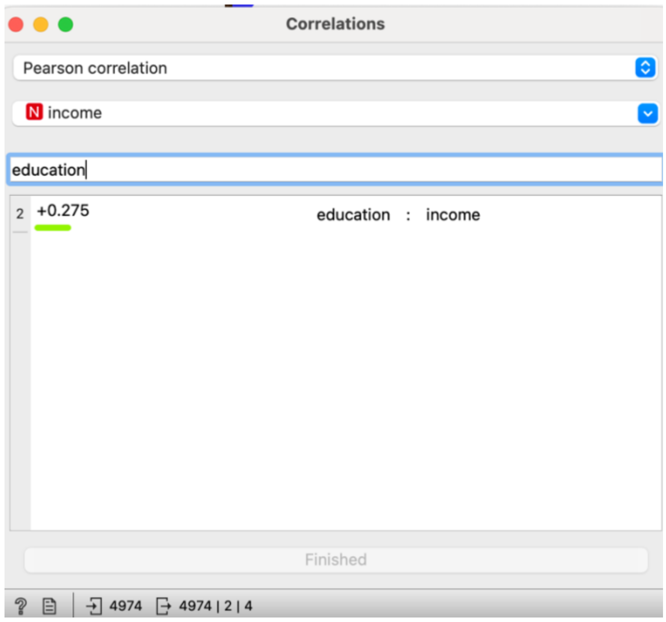


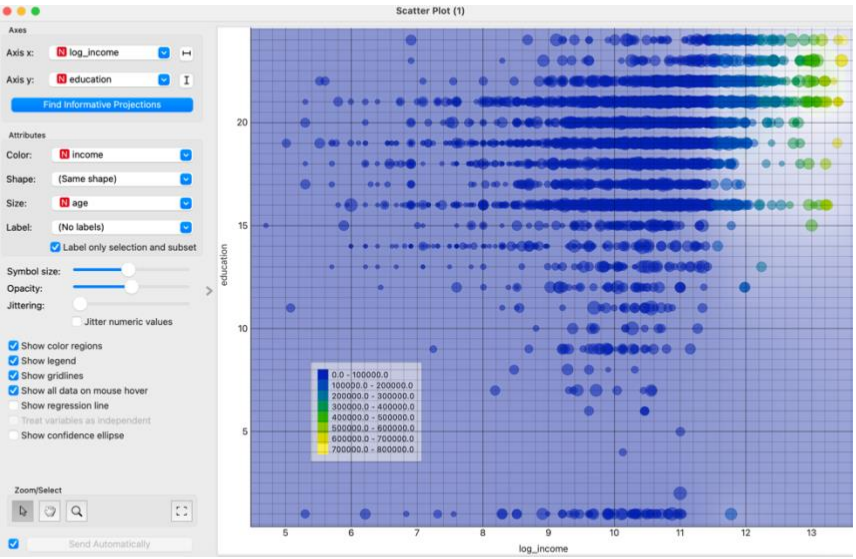


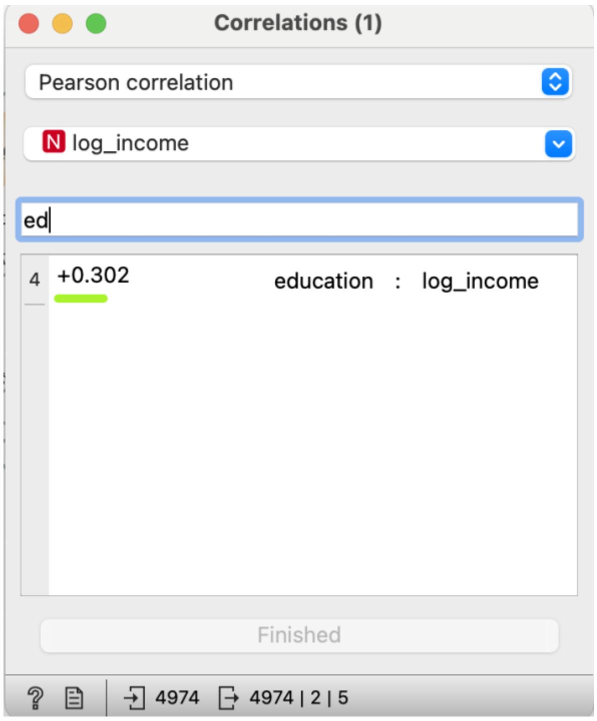


Income and education level graphs









Income Analysis

The analysis of income disparities reveals key insights across demographics. Males have higher median incomes and more high-income outliers than females, confirmed by a student's t-test ($t = 11.93$, $p < 0.001$, $N = 4974$). Scatterplots show males with more higher incomes, though factors like education and occupation should be considered.

For place of birth, income differences between US-born and non-US-born are minimal. US-born individuals have a slightly higher median income, but a t-test ($t = 0.840$, $p = 0.401$, $N = 4974$) shows no significant difference. Scatterplots suggest US-born individuals have a wider income distribution, though dataset imbalance limits generalisability.

Race analysis shows "White" individuals with higher median incomes and more outliers than "Non-White" individuals, confirmed by a t-test ($t = 5.004$, $p < 0.001$, $N = 4974$). Scatterplots visualise this disparity, but potential biases must be considered.

Income variation across age groups shows higher incomes among older adults, but exceptions exist. The Pearson correlation (+0.247) suggests a weak positive correlation, with other factors likely influencing income more. Log income transformation strengthens correlation (+0.496), showing a more pronounced age-income relationship.

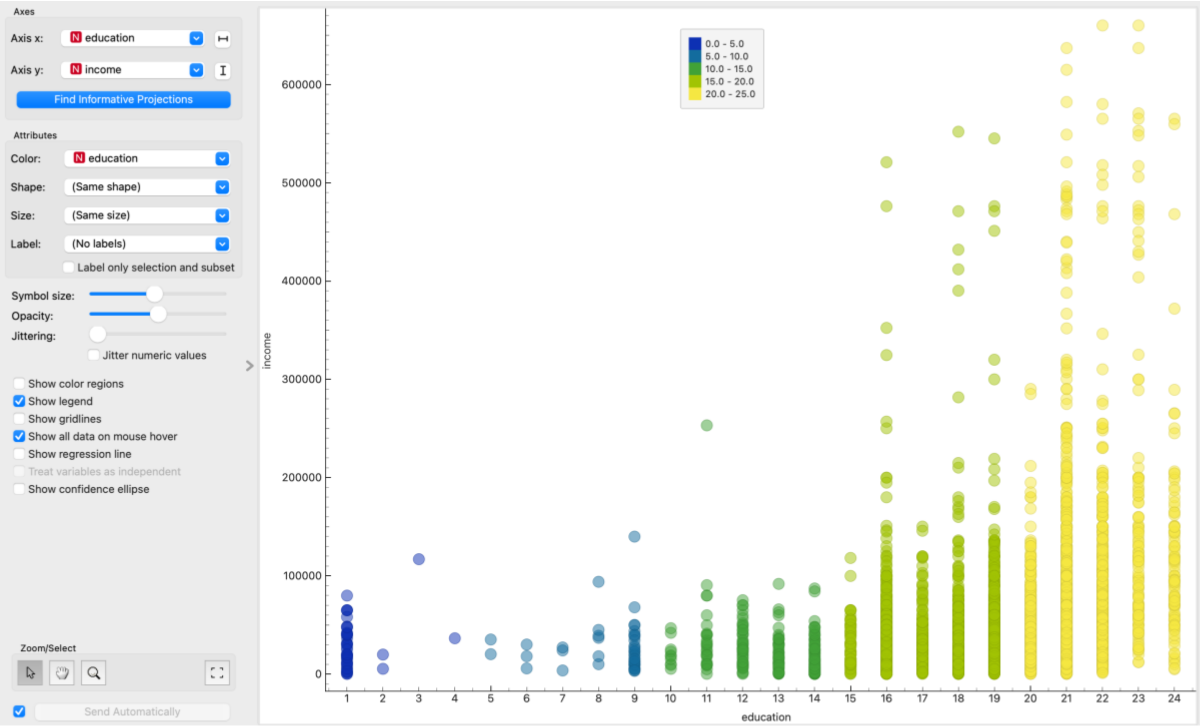
Income versus hours worked shows variability, with a Pearson correlation (+0.2968) indicating a weak positive link. Log transformation strengthens this correlation (+0.496).

Education level correlates with income, with a Pearson correlation (+0.275) indicating a weak positive link. Log transformation slightly strengthens this correlation (+0.302).

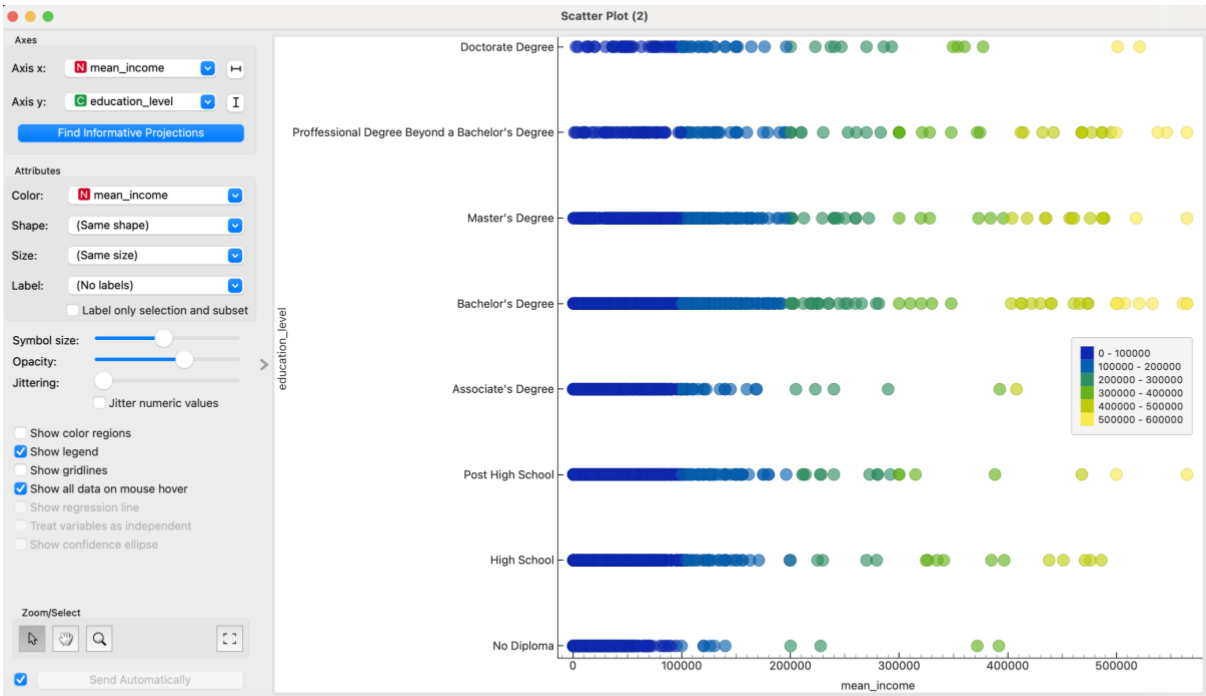
In summary, significant income disparities exist across sex, place of birth, and race, with males, US-born, and "White" individuals generally earning more. Age, hours worked, and education also influence income, but relationships are complex. Log transformations clarify trends by reducing outlier impact. Further analysis is needed for more robust conclusions.

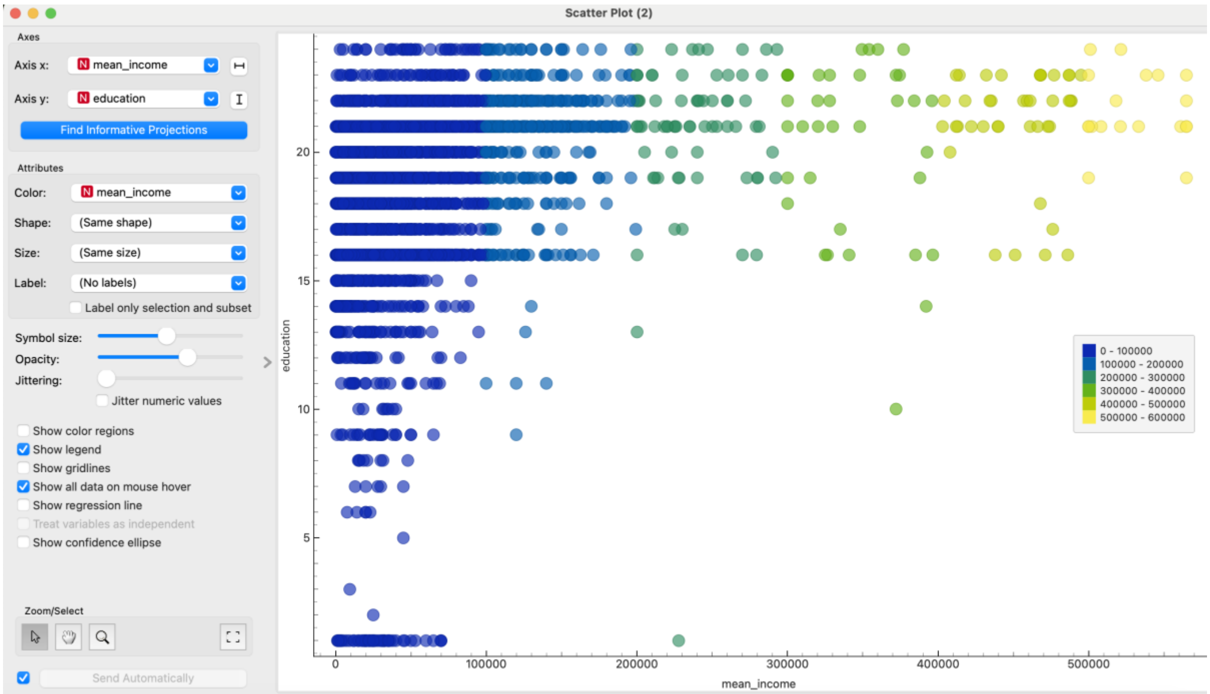
Part 3- Predicting Income

Education against income graph



Mean income against education level



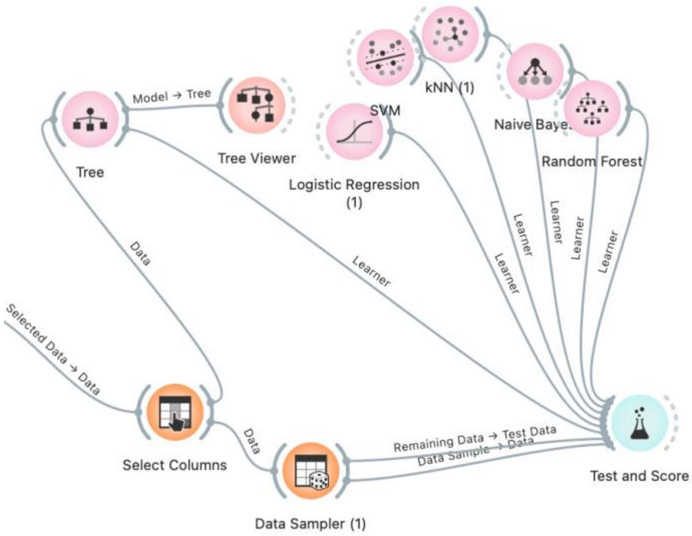


Scatter Plot (2)
Axes
Axis x: mean_income
Axis y: education
Find Informative Projections
Attributes
Color: mean_income
Shape: (Same shape)
Size: (Same size)
Label: (No labels)
Label only selection and subset
Symbol size:
Opacity:
Jittering:
Jitter numeric values
Show color regions
Show legend
Show gridlines
Show all data on mouse hover
Show regression line
Treat variables as independent
Show confidence ellipse
Zoom/Select
Send Automatically
education
mean_income
0 - 100000
100000 - 200000
200000 - 300000
300000 - 400000
400000 - 500000
500000 - 600000

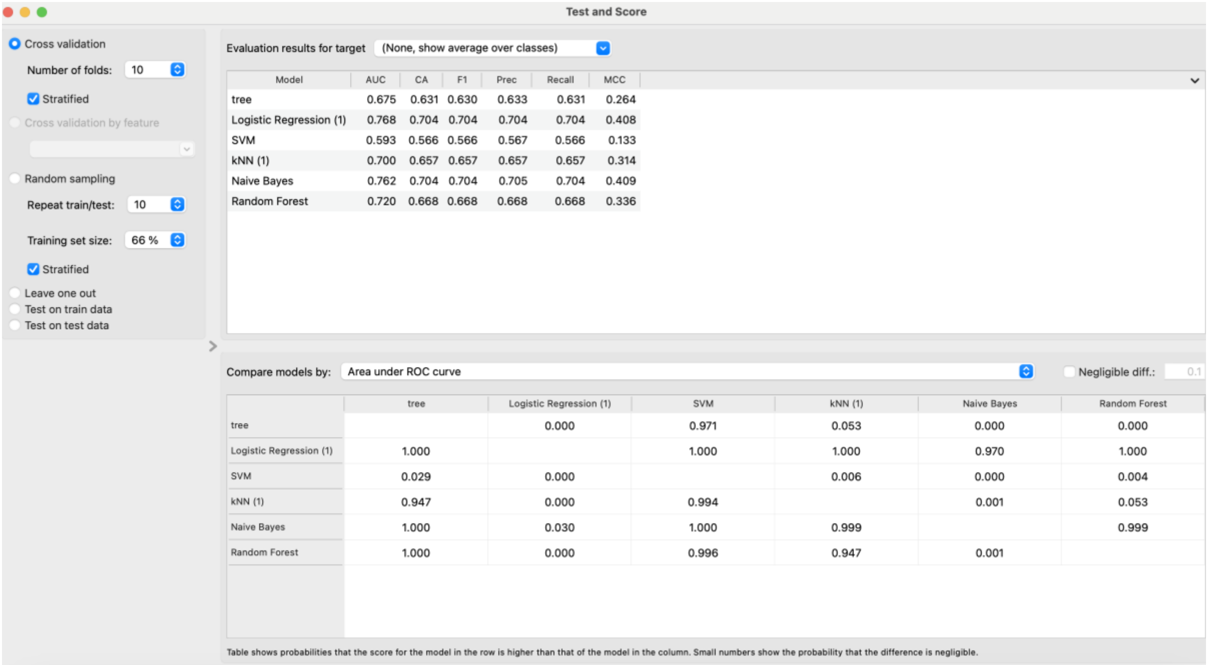
Analysis

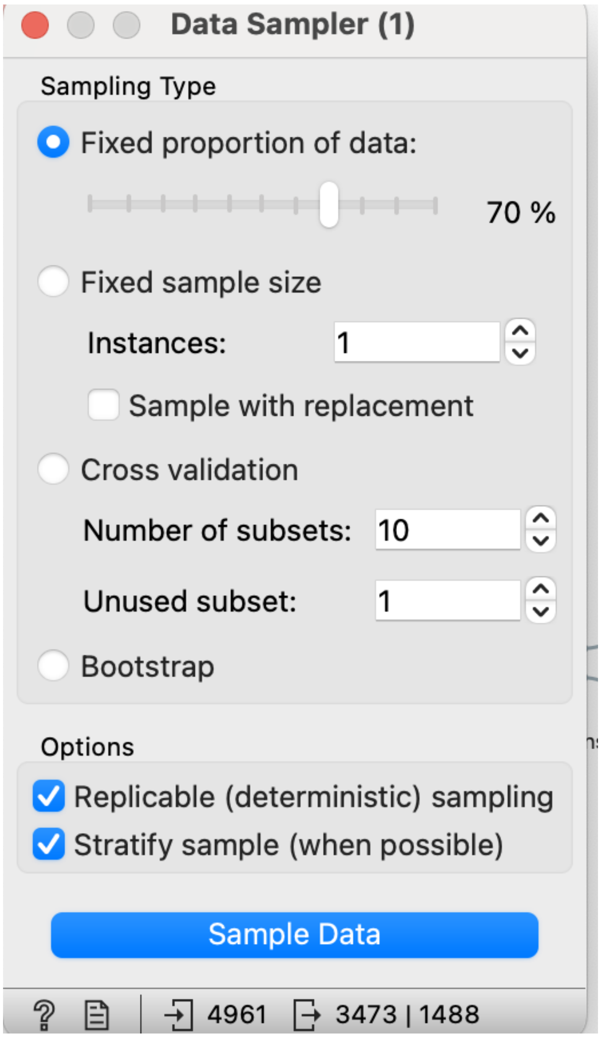
By using these scatterplots an estimate for the monetary value for education can be calculated, estimating how much one's income increases per year of education. At lower education levels (0-10), income remains consistently below $100,000, suggesting minimal monetary return per education level. A moderate increase, approximately $10,000 to $20,000 per level, appears between education levels 10-15, where income starts to spread towards $200,000. A substantial income jump, approximately $30,000 to $50,000 or more per level, is evident at higher education levels (15-20), with incomes distributed widely and exceeding $300,000. The highest incomes, above $400,000, are almost exclusively associated with the top education levels (20-24), indicating a significant monetary increase per level. However, this analysis is hindered by several factors. Confounding variables like field of study, experience, and location are not accounted for, potentially skewing the perceived impact of education. The plot demonstrates correlation, not causation, and visual estimates from the data distribution lack statistical rigor. Significant data overlap and potential outliers further limit the accuracy of income increase estimations per education level, making these figures approximate and requiring cautious interpretation.

Classification models



Before applying the classification models, I split the population into "low income" and "high income". I did this by looking at the median income value then creating the formula from this. I found 40,000 to be the median income. So, anything below this value was classified as "low income" and anything above this was classified as "high income". I then tried out several classifier models, using my testing and training data as seen by the data sampler, to predict which subgroup a person belongs to, this is seen by the different graphs coming from the test and score widget. I found that logistic regression and naïve bays had significantly better performance as seen by their test and score evaluations, as seen below. Classification accuracy (CA) measures the proportion of correctly classified instances, offering a straightforward view of the model's overall performance. In contrast, Area Under the ROC Curve (AUC) evaluates a model's ability to discriminate between classes across various thresholds, proving more robust to class imbalance and providing insights into the trade- off between true and false positive rates.


Test and Score
Cross validation
Number of folds: 10
Stratified
Cross validation by feature
Random sampling
Repeat train/test: 10
Training set size: 66 %
Stratified
Leave one out
Test on train data
Test on test data
Evaluation results for target (None, show average over classes)
Model AUC CA F1 Prec Recall MCC
tree 0.675 0.631 0.630 0.633 0.631 0.264
Logistic Regression (1) 0.768 0.704 0.704 0.704 0.704 0.408
SVM 0.593 0.566 0.566 0.567 0.566 0.133
kNN (1) 0.700 0.657 0.657 0.657 0.657 0.314
Naive Bayes 0.762 0.704 0.704 0.705 0.704 0.409
Random Forest 0.720 0.668 0.668 0.668 0.668 0.336
Compare models by: Area under ROC curve
Negligible diff.: 0.1
tree Logistic Regression (1) SVM kNN (1) Naive Bayes Random Forest
tree 0.000 0.971 0.053 0.000 0.000
Logistic Regression (1) 1.000 1.000 1.000 0.970 1.000
SVM 0.029 0.000 0.006 0.000 0.004
kNN (1) 0.947 0.000 0.994 0.001 0.053
Naive Bayes 1.000 0.030 1.000 0.999 0.999
Random Forest 1.000 0.000 0.996 0.947 0.001
Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

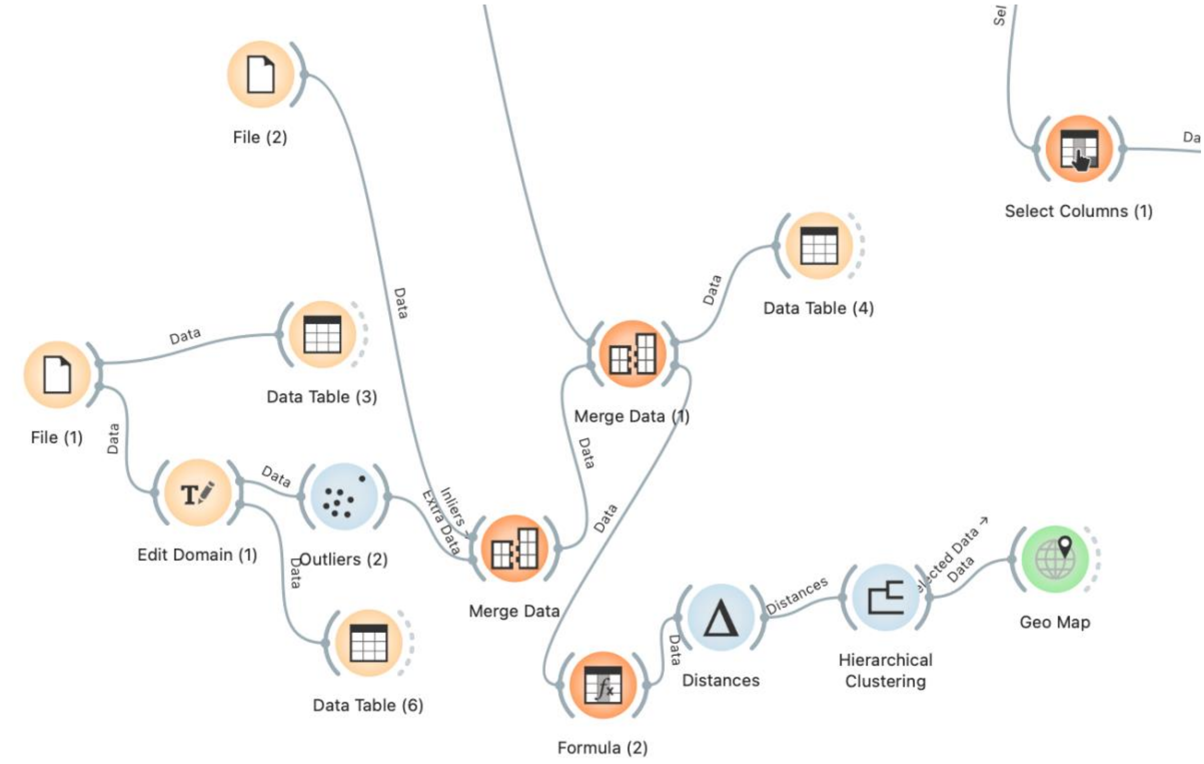

Data Sampler (1)
Sampling Type
Fixed proportion of data:
70 %
Fixed sample size
Instances: 1
Sample with replacement
Cross validation
Number of subsets: 10
Unused subset: 1
Bootstrap
Options
Replicable (deterministic) sampling
Stratify sample (when possible)
Sample Data
4961
3473 | 1488

Feature ranking using income as the target results

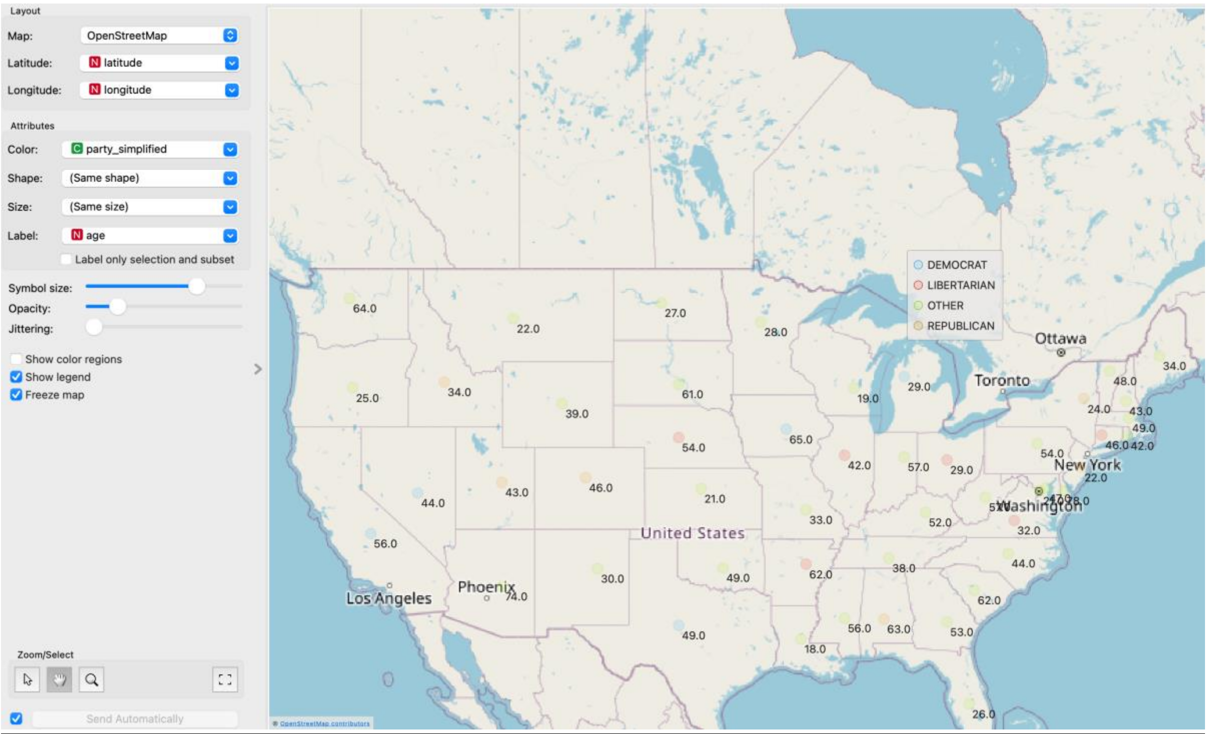
		#	Info. gain	Gain ratio	Gini	χ²	ReliefF	FCBF
1	N income		1.000	0.500	0.500	3259.958	0.045	2.000
2	C sex	2	0.018	0.018	0.012	62.991	0.020	0.018
3	C race	2	0.006	0.008	0.004	33.834	0.004	0.007
4	N hours		0.148	0.083	0.096	670.832	0.010	0.000
5	N occupation		0.082	0.041	0.054	250.745	0.020	0.000
6	N education		0.080	0.041	0.053	414.586	0.007	0.000
7	N age		0.075	0.037	0.050	251.398	0.007	0.000
8	C Education Type	8	0.090	0.034	0.059	619.167	0.058	0.000
9	C marital	5	0.061	0.039	0.041	860.126	0.010	0.000
10	C state	51	0.013	0.003	0.009	9.947	-0.002	0.000
11	N PoB		0.002	0.001	0.001	0.121	0.008	0.000
12	C us_born	2	0.001	0.001	0.000	0.646	0.002	0.000
13	C CoW	10	0.013	0.007	NA	36.287	0.052	NA

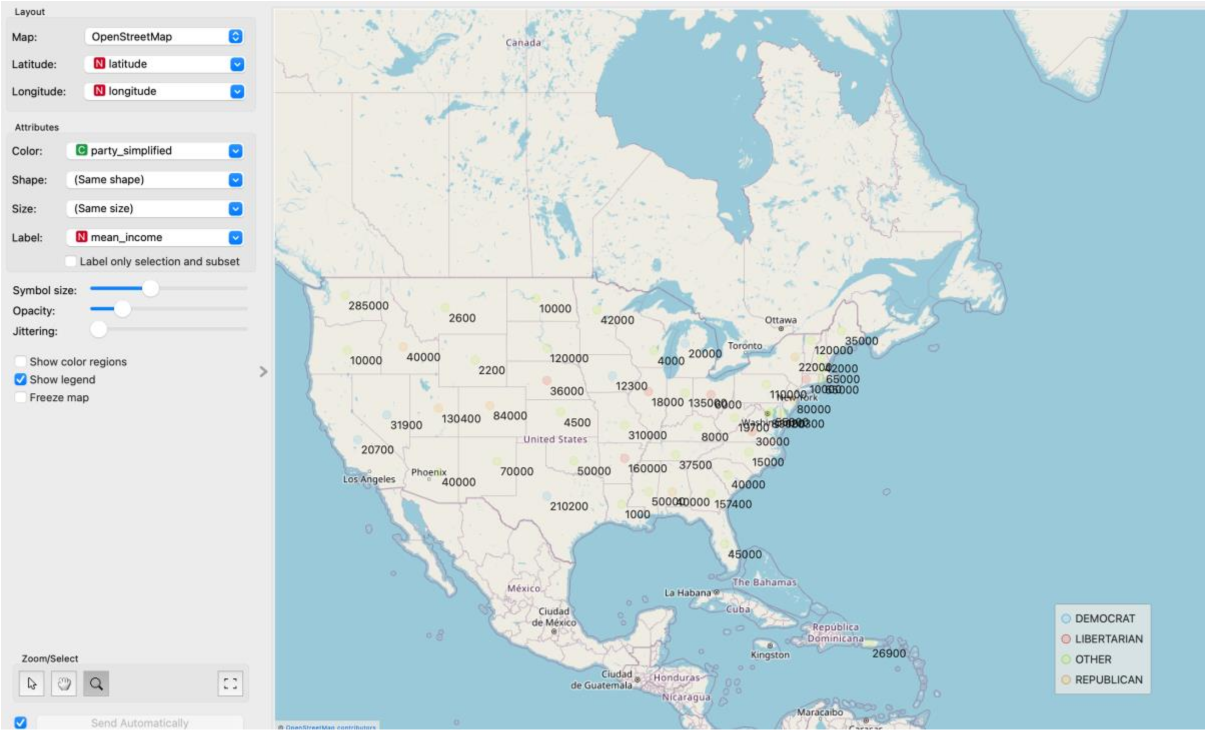
The feature importance analysis aims to identify the most influential factors in predicting income. The results indicate that work-related factors such as hours and occupation along with education related factors are strong predictors of income, while demographic factors like race, place of birth, appear to be less influential. This therefore suggests that features related to an individual's skills, occupation, and educational background are key determinants of income.

Part 4- Demographics of US elections



State maps produced- age, mean income, mean education




Layout
Map:
OpenStreetMap
Latitude:
latitude
Longitude:
longitude
Attributes
Color:
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Shape:
(Same shape)
Size:
(Same size)
Label:
mean_income
Label only selection and subset
Symbol size:
Opacity:
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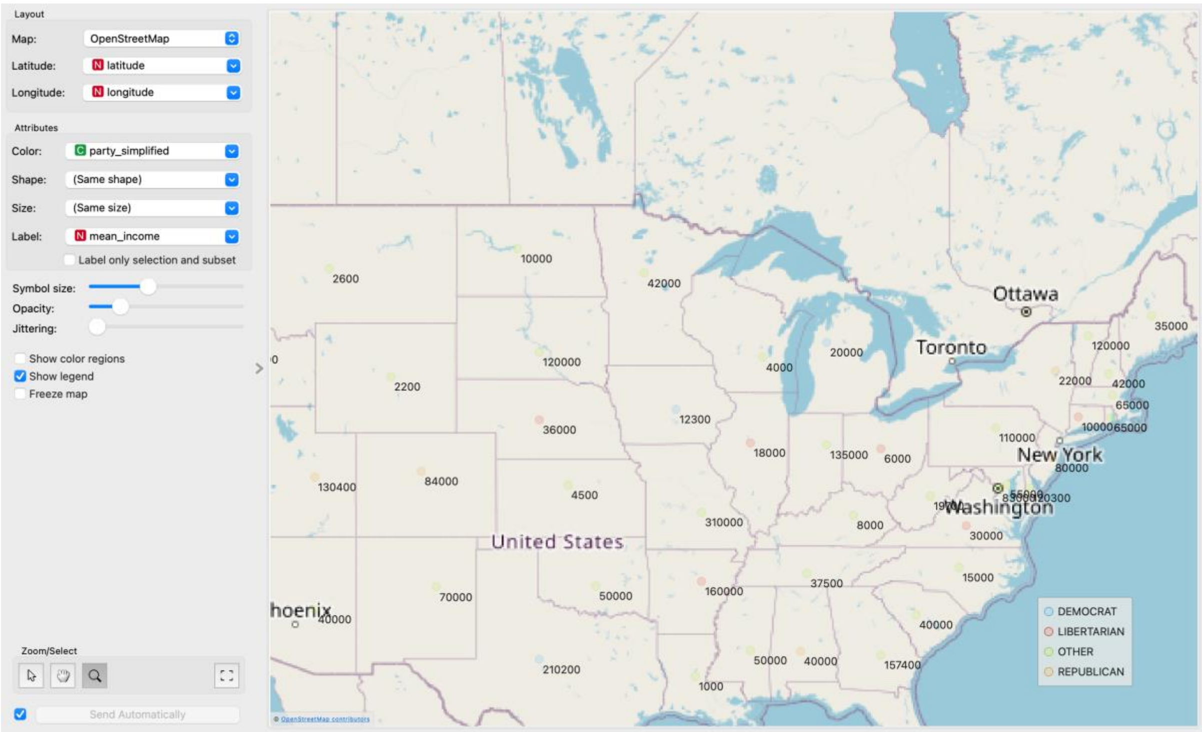
Show color regions
Show legend
Freeze map
Zoom/Select
Send Automatically
DEMOCRAT
LIBERTARIAN
OTHER
REPUBLICAN


Layout
Map:
OpenStreetMap
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latitude
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longitude
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party_simplified
Shape:
(Same shape)
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(Same size)
Label:
mean_income
Label only selection and subset
Symbol size:
Opacity:
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Zoom/Select
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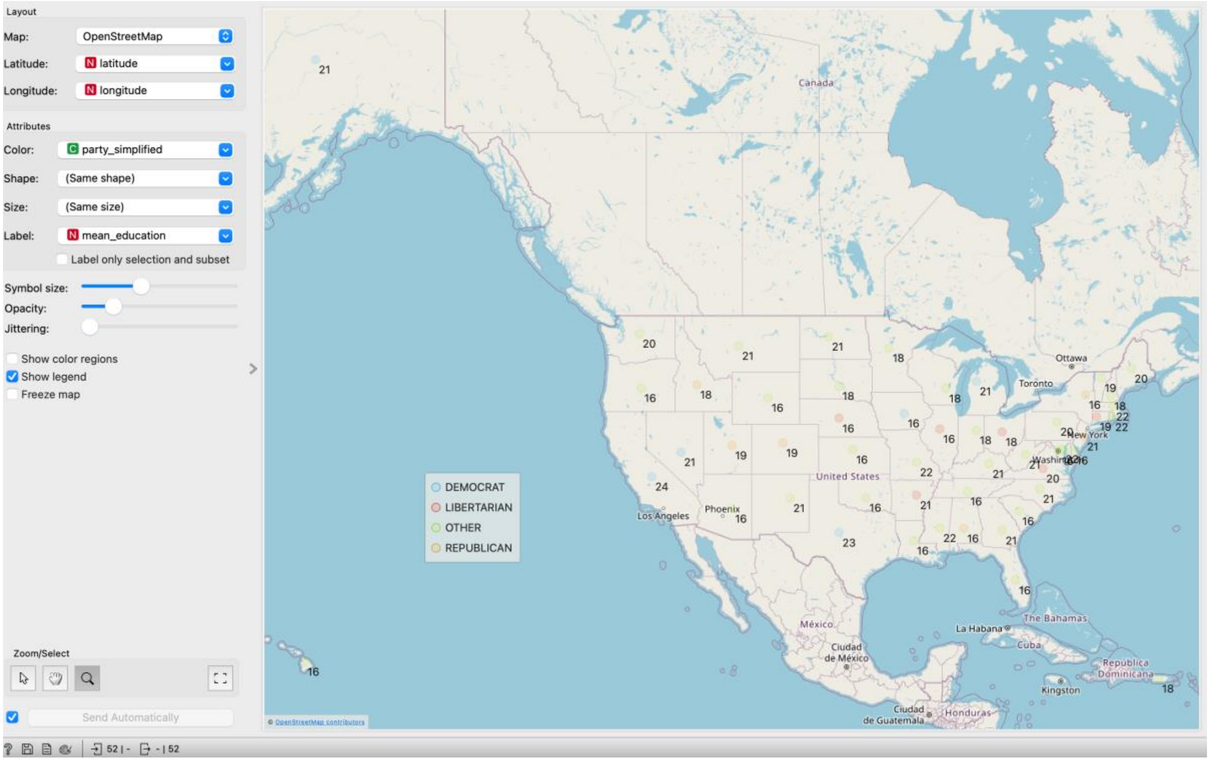
OTHER
REPUBLICAN


Layout
Map: OpenStreetMap
Latitude: latitude
Longitude: longitude
Attributes
Color: party_simplified
Shape: (Same shape)
Size: (Same size)
Label: mean_education
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Symbol size:
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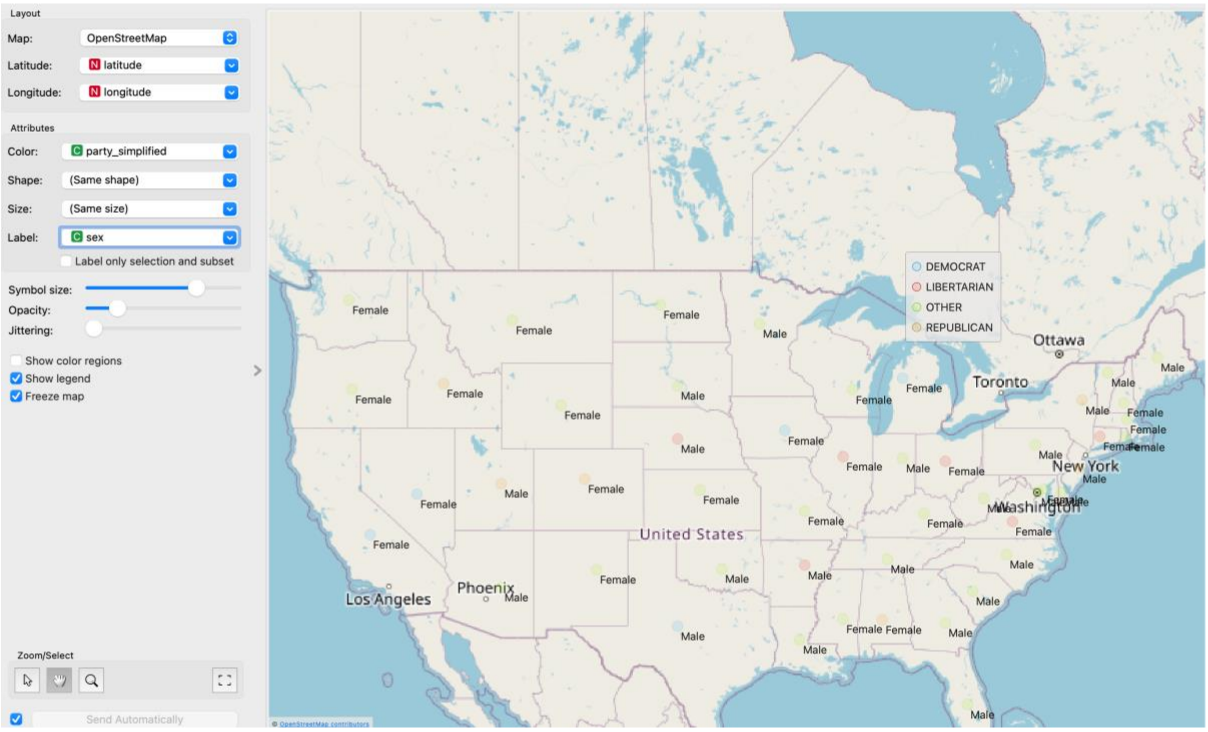
Jittering:
Show color regions
Show legend
Freeze map
Zoom/Select
Send Automatically
DEMOCRAT
LIBERTARIAN
OTHER
REPUBLICAN
Canada
United States
Ottawa
Toronto
New York
Washington
Phoenix
Los Angeles
México
Ciudad de México
La Habana
Cuba
The Bahamas
República Dominicana
Kingston
Ciudad de Guatemala
Honduras
52 | - | 52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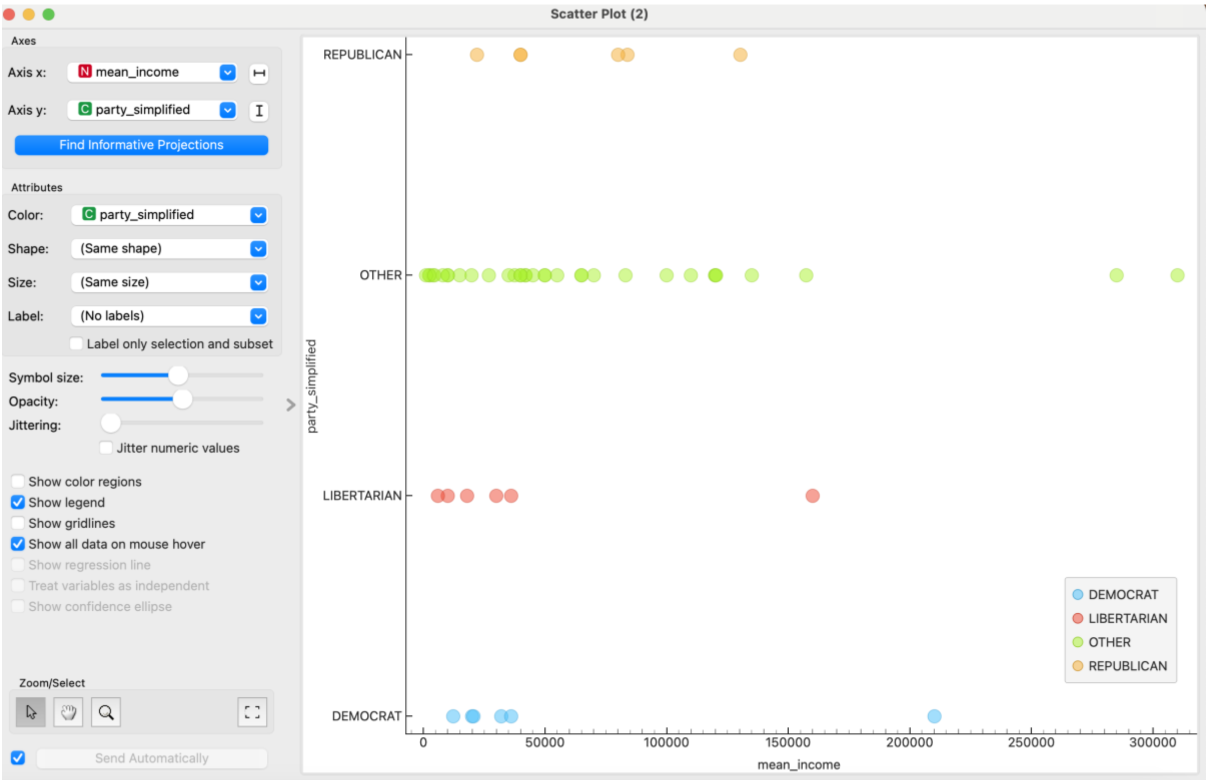
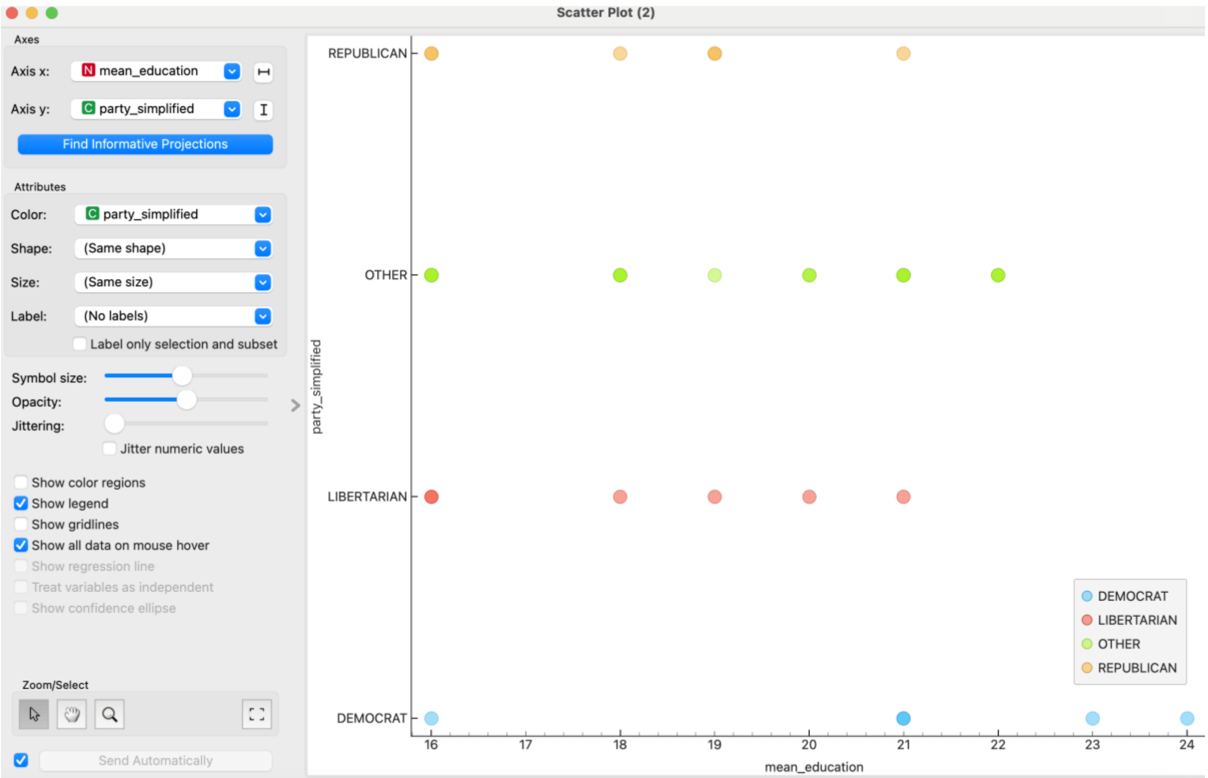
Label: sex
Label only selection and subset
Symbol size:
Opacity:
Jittering:
Show color regions
Show legend
Freeze map
Zoom/Select
Send Automatically
DEMOCRAT
LIBERTARIAN
OTHER
REPUBLICAN
Ottawa
Toronto
New York
Washington
United States
Phoenix
Los Angeles

Scatterplot of mean income





Analysis

For this part, I used a dataset called "voting_2020.csv," which contains the election results by state for the 2020 Trump vs. Biden election. Before creating the geo maps, I ensured that the state columns in both datasets were in the same format and merged them accordingly. I again edited the data columns using the edit widget and removed any outliers. Additionally, I downloaded a longitude and latitude dataset of U.S. states and merged this data using the state column. Using hierarchical clustering, I then created maps based on the different state clusters. Hierarchical clustering allows natural groupings or clusters within the data, which can reveal underlying patterns that may not be immediately apparent through other methods.

From the U.S. map, it is evident that a significant number of people voted for parties other than Trump and Biden. In general, more females than males voted, and both Democratic and Republican parties had the majority of their votes coming from a higher age demographic.

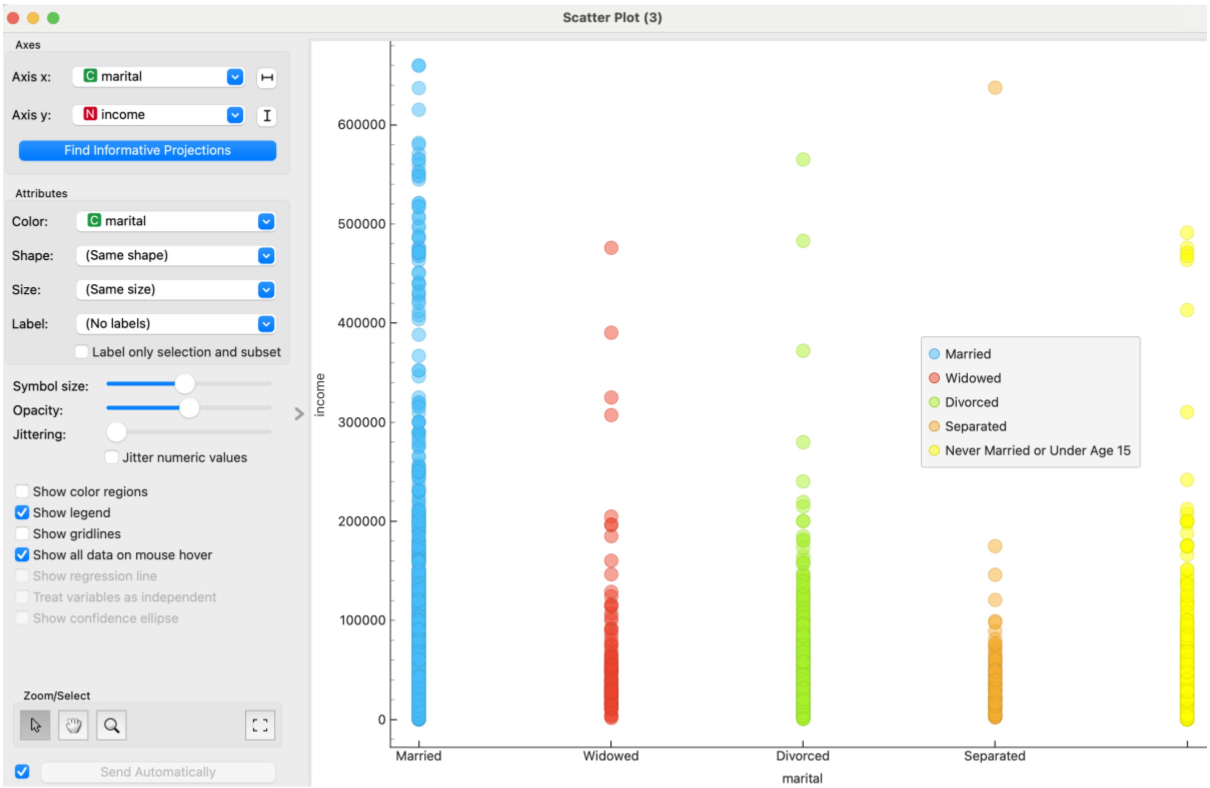
Analysing the U.S. map showing mean income, it appears that higher incomes tended to vote for Trump (Republican). However, there was one state with a particularly high income of $210,200 that voted for Biden (Democratic). The U.S. map displaying mean educational attainment levels indicated that states voting for Biden generally had higher educational levels. States voting for Trump were not particularly low, with an average educational attainment around 18/19, which corresponds to some college education but not a degree.

To gain a more accurate visualisation, I conducted deeper analysis using scatterplots. The scatterplot of mean income shows that Trump-voting states are spread across various income levels, including lower income brackets. This supports the hypothesis that some low-income states may vote Republican. However, there are also Republican states at higher income levels, suggesting a varied income range. In contrast, Biden-voting states are primarily concentrated at lower income levels, with one exception at a very high-income level.

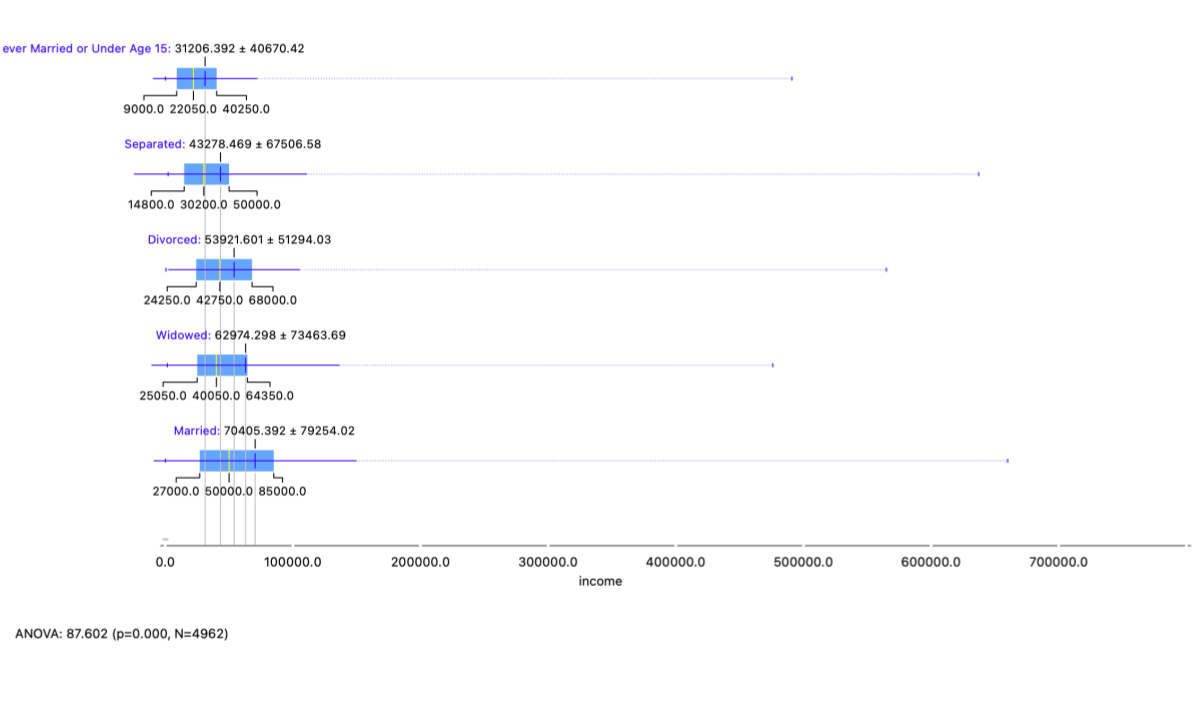
Additionally, I examined the scatterplot for mean education to test the hypothesis that states with high educational attainment voted for Biden. The scatterplot reveals that Democrat-voting states are predominantly at higher educational levels, with mean education ranging from 21 to 24. This distribution supports the hypothesis. Republican-voting states are distributed across the lower spectrum of educational attainment, indicating that lower educational attainment may be more inclined to vote Republican (Trump). However, while the analysis shows correlations it is important to understand these may not be causations and there may be bias/ cofounding variables that are also influencing these results. Such as age and their occupation type.

Part 5- My Data Mining

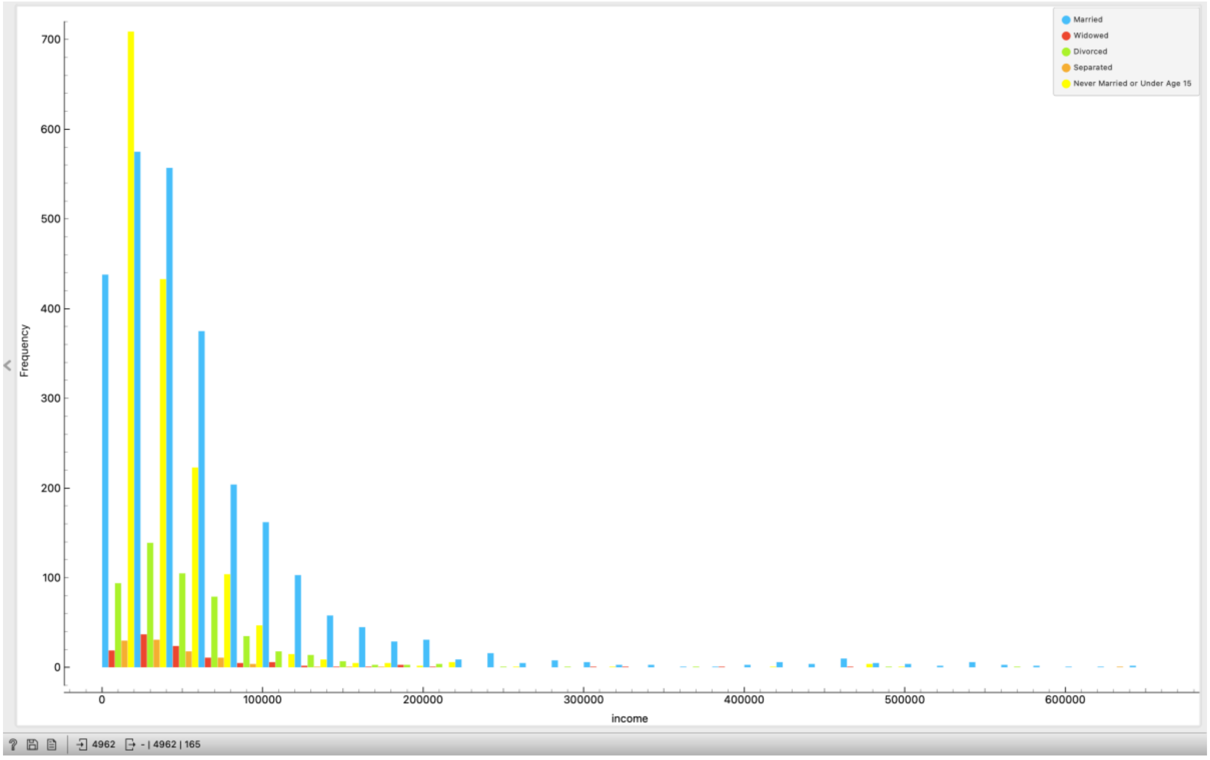
Scatterplot



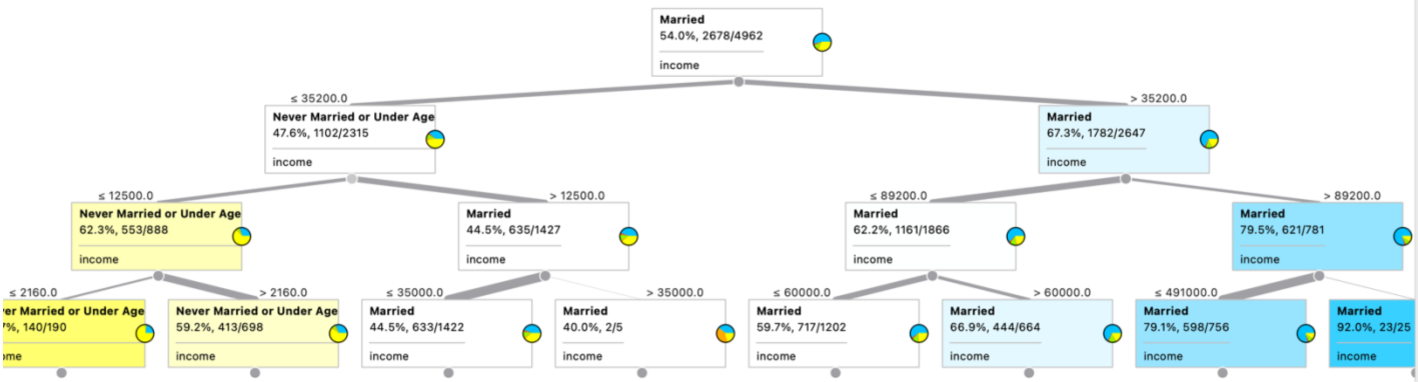
Boxplot



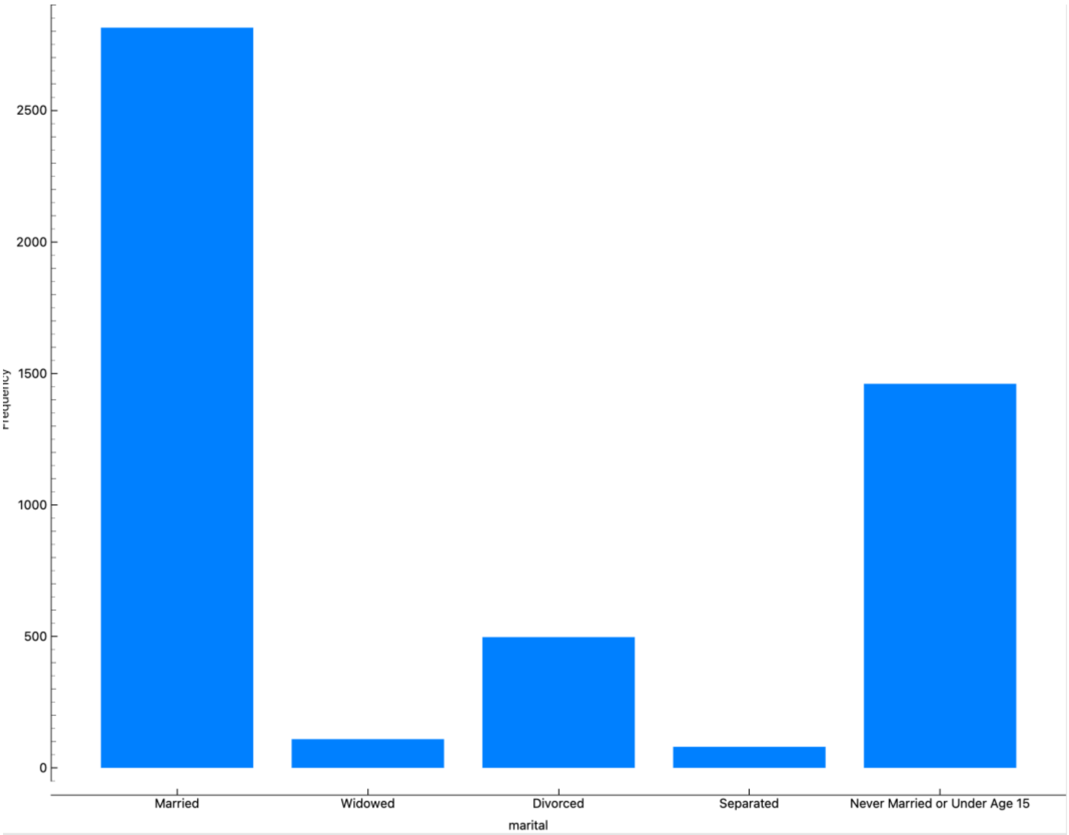
Distribution Plot



Decision Tree



Marital Distribution



Analysis

In this section of the coursework, I will analyse the hypothesis that "married people are more likely to be on a higher income".

Scatterplot Analysis:

Initially, I examined the scatterplot of marital status against income. The data reveals that married individuals exhibit a broad income distribution, with many reaching higher income levels. This observation supports the hypothesis that married people are more likely to have higher incomes.

Boxplot Analysis:

Next, I analysed the boxplot for marital status and income. The boxplot indicates that the median income for the married category is the highest at $50,000. The interquartile range (IQR) shows the spread of the middle 50% of the data, with the "married" category positioned further to the right on the income scale compared to others, indicating a generally higher income distribution. The whiskers extend further to the right, suggesting a wider range of higher incomes. The mean income for married individuals is also the highest, reinforcing the hypothesis. An ANOVA test provides strong statistical evidence that mean income significantly varies across marital categories, with a large F-statistic indicating substantial variation and a p-value of 0.000 demonstrating these differences are highly unlikely to be due to chance. With a sample size of 4,962, the test exhibits high statistical power, supporting the conclusion that marital status is significantly associated with income.

Distribution Plot Analysis:

The distribution plot visually supports the hypothesis, showing that married people are more likely to have higher incomes. While all marital status categories concentrate in lower income ranges, the "married" category extends further into higher brackets, indicating a greater prevalence of higher earners. In contrast, the "never married or underage 15" category is strongly concentrated in lower income ranges, suggesting that married individuals are more likely to span a wider spectrum of income levels, including the higher end.

Decision Tree Analysis:

Further analysis using a decision tree provides significant visual evidence supporting the hypothesis. The tree structure reveals a clear trend: as income levels increase, the proportion of married individuals rises significantly, reaching 92% in the highest income bracket. Conversely, unmarried individuals, particularly those "never married or under age 15," are more prevalent at lower income levels. The tree splits based on income thresholds, such as an initial split at $35,200, highlighting the income ranges where marital status effectively differentiates income levels. However, it is important to note that while the tree illustrates an association between marital status and income, it represents correlation, not causation. Other factors, such as age and culture may influence these outcomes.

Important Considerations

Although this analysis agrees with the hypothesis “married people are more likely to be on a higher income”, it is important to understand that the data in this analysis included significantly more married people than any other marital status, as seen by the marital distribution plot. This could therefore cause an inaccurate representation. Furthermore, as stated within the tree analysis other factors such as age and culture may influence these outcomes.